

Tsunami-Surrogate: A Controlled Multi-Reference Benchmark for Neural-Operator Emulation of Tsunami-Like Wave Propagation

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July 4, 2026

Abstract

We study neural-operator surrogates for tsunami-like wave propagation in a controlled synthetic benchmark based on two-dimensional shallow-water and weakly dispersive reference simulations. Each scenario pairs a procedurally generated bathymetry field with a synthetic initial displacement source, and the same scenario can be advanced by multiple numerical references to separate surrogate error from reference-choice effects. Under matched train/test protocols, the primary experiments evaluate FNO and factorized FNO (FFNO) surrogates against convolutional and operator baselines while measuring solver-gap diagnostics, implementation-level inference speedup, distribution-shift behavior through metadata-filtered and strict held-out-family diagnostics, cross-resolution transfer, physics diagnostics, and ensemble uncertainty with $M = 7$ members. We also test whether models trained on the synthetic distribution can be applied beyond the procedural terrain generator by evaluating direct and seeded windowed rollouts on a fully wet, 10-crop real-bathymetry morphology suite under synthetic forcing. These transfer results are interpreted conservatively: they probe morphology shift under controlled forcing, not real-event validation, operational tsunami warning, or site-specific hazard assessment.

1. Introduction

Physics-based simulation of tsunami-like wave propagation can become expensive when repeated across many possible source scenarios, bathymetric fields, numerical resolutions, or uncertainty-analysis runs. This cost motivates surrogate models that approximate numerical rollouts more rapidly while retaining the spatial and temporal structure needed for scientific evaluation.

We cast forward emulation as an operator-learning problem: given a bathymetry field and an initial free-surface disturbance, the model predicts a future free-surface trajectory generated by a numerical reference solver. Fourier Neural Operators (FNOs) are a natural primary model because they learn mappings between discretized functions and can be evaluated on spatial fields. We also include a factorized FNO (FFNO), mode-count ablations, U-FNO and WNO variants, while CNN and U-Net baselines provide simpler convolutional comparisons.

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The goal is not to replace site-specific tsunami models or to claim operational early-warning readiness. Instead, we study a narrower scientific question: under controlled synthetic conditions, can neural surrogates provide useful speed–accuracy trade-offs for solver-generated tsunami-like wavefields, and how do those trade-offs change under distribution shift, resolution transfer, solver mismatch, predictive uncertainty, and transfer from synthetic to real bathymetry?

The main contributions of this study are:

1. A controlled forward benchmark that pairs procedurally generated bathymetry with diverse initial-displacement sources and shared numerical scenarios,
2. a multi-reference label design with hydrostatic reconstruction as the primary reference, MUSCL-HR as a second-order comparison reference, and Boussinesq as a weakly dispersive comparison reference on the same scenarios,
3. an FNO/FFNO-centered surrogate study with CNN, U-Net, U-FNO, WNO, and Fourier-mode ablations, together with seeded windowed FNO and FFNO variants for long-horizon rollout diagnostics,
4. an evaluation suite covering reference-solver health checks, speed, metadata-filtered and strict held-out-family OOD diagnostics, cross-resolution behavior, solver-vs-solver analysis, emulator-superiority-style ratios, physics diagnostics, ensemble uncertainty, and a synthetic-source real-bathymetry morphology suite.

2. Related Work

Tsunami and shallow-water numerical solvers. Tsunami propagation is commonly modeled with depth-averaged shallow-water equations, where finite-volume methods and Riemann-solver-based fluxes provide a robust framework for wave propagation over nonuniform bathymetry [16, 28]. Operational and research tsunami codes such as GeoClaw demonstrate the importance of well-balanced finite-volume methods, wet–dry handling, and adaptive refinement for transoceanic propagation and coastal inundation [17, 2]. In our benchmark, the primary reference solver follows the hydrostatic-reconstruction idea of Audusse et al. [1], which preserves the lake-at-rest equilibrium and nonnegative water depth. We additionally include a second-order MUSCL-HR reference motivated by finite-volume tsunami simulation with hydrostatic reconstruction [5], while using Rusanov-type numerical fluxes and wave-speed estimates in the spirit of standard hyperbolic finite-volume methods [28, 29]. We also include a Boussinesq-type weakly dispersive comparison reference, since Boussinesq-type solvers provide context for modeling dispersive wave effects beyond non-dispersive shallow-water theory [25, 4].

Synthetic bathymetry and source generation. Because the goal of this work is a controlled benchmark rather than a site-specific operational model, we procedurally generate bathymetry and initial disturbance fields. Procedural terrain modeling and fractal/noise-based geometry generation have long been used to create diverse spatial fields [20]. We use these ideas only as a controllable source of synthetic bathymetric variability, not as a replacement for real bathymetric products. The source generator includes simple Gaussian, dipole, rough, and Okada-like families; the latter

is motivated by classical elastic half-space deformation models for earthquake sources [21]. Real global marine gravity and bathymetry products [24] are relevant to future site-specific extensions, but this paper intentionally focuses on synthetic scenarios for controlled generalization tests.

Neural operators and baseline architectures. Neural operators learn maps between function spaces and are now standard tools for PDE surrogate modeling [18, 13]. We use FNO as the main neural-operator surrogate and compare it with simpler convolutional baselines such as CNNs and U-Nets [15, 23]. Later operator variants include factorized Fourier operators, U-FNO-style multiscale Fourier/U-Net hybrids, and wavelet neural operators, which motivate our baseline coverage rather than a broad architecture survey [30, 32, 31]. Our contribution is not a new architecture; instead, we evaluate established surrogate families under a controlled multi-reference tsunami-like benchmark. Recent neural-operator work on localized kernels and geophysical autoregressive forecasting further motivates evaluating operator models under local structure, multi-step forecast horizons, and resolution changes [19, 22, 10].

Tsunami-oriented neural surrogates. Machine-learning approaches have recently been explored for tsunami propagation, inundation, and rapid assessment. Closest to our setting, Kim et al. [11] propose a Factorized Fourier Neural Operator surrogate for basin-scale tsunami propagation using COMCOT-generated wavefields, held-out source scenarios, arrival-time errors, and speed comparisons. PINN and DeepONet-style models have also been studied for tsunami inundation modeling [3], while neural-network surrogates have been used to accelerate probabilistic tsunami inundation assessment [6]. Recent work has also considered neural operators for rapid tsunami propagation and source inversion [26]. These studies motivate neural surrogates for tsunami-related modeling, but our contribution is different: we construct a controlled benchmark with synthetic bathymetry/source generators, multiple shallow-water numerical references, explicit OOD and cross-resolution protocols, solver-vs-solver analysis, and uncertainty evaluation.

Scientific machine-learning benchmarks. Our work is also related to benchmark-oriented scientific machine learning. PDEBench provides a broad benchmark suite for time-dependent PDE problems, including generated datasets, baseline models, and multiple evaluation metrics [27]. In contrast, our benchmark is narrower but domain-focused: we target tsunami-like shallow-water propagation with controlled bathymetry/source variability and multi-solver reference labels.

Solver-gap and emulator-superiority-style evaluation. A central difficulty in PDE surrogate benchmarking is that the training solver is not necessarily ground truth. Recent work on neural emulator superiority shows that an emulator trained on lower-fidelity numerical data can, in specific regimes, be closer to a higher-fidelity reference than the solver that generated its training data [12]. This motivates our multi-reference evaluation: we report same-solver surrogate error, raw solver-vs-solver gaps, cross-solver surrogate evaluation, and emulator-superiority-style ratios only when the normalization and reference choices are scale-compatible. This separates surrogate approximation error from differences among numerical reference solvers.

Uncertainty quantification. For hazard-oriented surrogate models, point accuracy alone is insufficient; users also need to know when the model may be unreliable. Deep ensembles provide a simple and scalable approach to predictive uncertainty estimation and often give strong calibration without major changes to the base model [14]. We therefore use ensembles as the primary uncertainty mechanism and evaluate whether uncertainty correlates with actual error and increases under OOD scenarios. Broader surveys and Bayesian deep-learning perspectives provide context for interpreting epistemic and aleatoric uncertainty in deep models [8, 33].

Inverse problems. Inverse source reconstruction is an adjacent direction, with classical adjoint inversion and recent neural inverse-operator methods providing relevant foundations. We leave this task to follow-up work and focus here on forward emulation and evaluation.

3. Data generation pipeline

3.1. Benchmark design

The dataset is built around shared synthetic scenarios rather than solver-specific samples. Each scenario contains a procedurally generated bathymetry field, a synthetic initial-displacement source field, and metadata recording the sampled bathymetry family, source family, random strength factor, and file locations. Numerical reference trajectories are then produced by advancing the same scenario with one or more forward solvers. This separation between scenario generation and reference generation is important for multi-reference evaluation: when two solver labels differ, the difference is attributable to the numerical reference and not to a different random seabed or initial disturbance. Figure 1 shows representative accepted scenario pairs spanning the morphology and source families described below.

The default forward benchmark uses a uniform 64×64 Cartesian grid on normalized coordinates, with matched bathymetry, source, and solver resolution. The training, validation, and test data come from three independently generated pools of 10,000, 1,000, and 2,500 scenarios (13,500 in total), each produced by a separate generation run with its own fixed per-pool seed rather than by partitioning a single pool, so no scenario is shared between pools by construction; the processed splits are described in Section 3.5. Each trajectory is advanced for 250 solver updates, and states with solver-step indices $0, 5, 10, \dots, 250$ are stored, giving 51 frames per rollout. Automatic time stepping is enabled with a target CFL value of 0.45 for the shallow-water references, so these saved frames are rollout snapshots with recorded timestamps and time-step histories rather than values interpolated onto a common physical-time grid. The default solver set now contains three references on the same scenario IDs: a hydrostatic-reconstruction shallow-water solver, a second-order MUSCL-HR shallow-water solver, and an elevation-only Boussinesq solver. Hydrostatic reconstruction remains the primary reference, while MUSCL-HR and Boussinesq provide two different comparison baselines for solver-gap analysis.

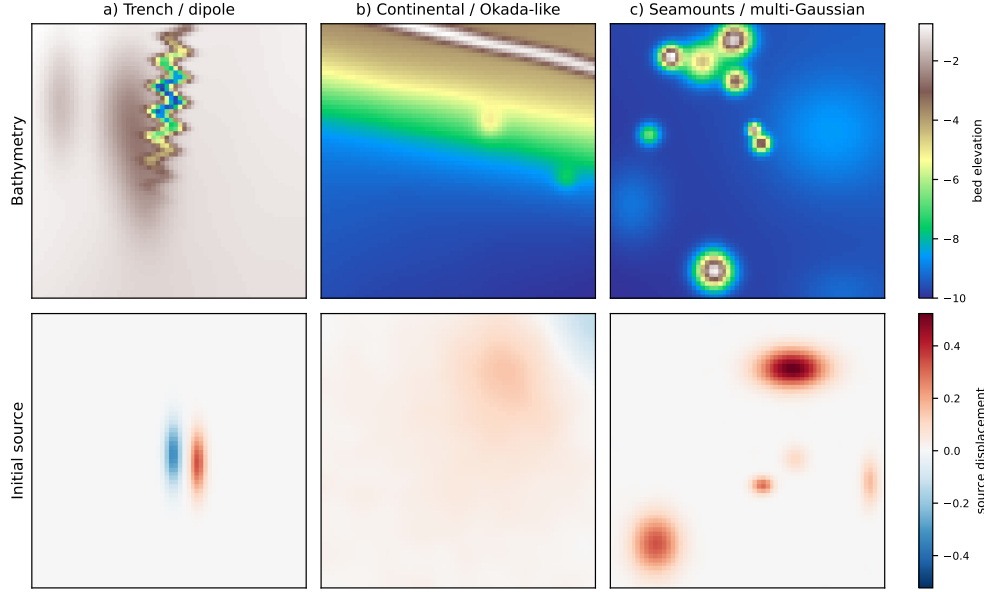


Figure 1: Representative accepted synthetic scenarios from the benchmark generator. Each column shows one paired bathymetry/source scenario before rollout: trench/dipole, continental/Okada-like, and seamounts/multi-Gaussian. Bathymetry uses the shared fully wet range $[-10, -0.75]$; source fields retain family-dependent amplitudes and are shown on a shared symmetric uplift/subsidence scale.

The benchmark is intentionally synthetic. Its purpose is to create a controlled supervised-learning problem for testing fast forward emulation of solver-generated tsunami-like wavefields, not to reproduce a specific coastline or to provide operational hazard assessment. This design makes it possible to evaluate solver-gap behavior, inference speed, distribution-shift behavior, cross-resolution transfer, and uncertainty quality under repeatable conditions.

3.2. Procedural bathymetry

Bathymetry is represented by a signed bed-elevation field $b(x, y)$, where $b(x, y) \leq 0$ denotes submerged cells below mean sea level. In the default configuration, every generated field is mapped into the range

$$[d_{\min}, d_{\max}] = [-10, -0.75].$$

This keeps the whole computational domain wet. The corresponding still-water depth is computed as

$$h_{\text{rest}}(x, y) = \max\{-b(x, y) + \eta_{\text{sea}}, 0\},$$

where η_{sea} is a configurable sea-level offset, set to zero in the default experiments.

For each scenario, the bathymetry generator samples one of five terrain families:

$$\{\text{trench}, \text{continental}, \text{seamounts}, \text{canyon}, \text{island}\}.$$

The generator first creates a low-frequency base field. In the current default configuration, this base is a weak rotated slope with slope components sampled from $[0.02, 0.12]$, although flat and basin-like base modes are also supported. To reduce grid-aligned artifacts, the normalized coordinates are smoothly warped before the terrain-family morphology is evaluated:

$$x_w = \text{clip}(x + \alpha g_x(x, y), 0, 1), \quad y_w = \text{clip}(y + \alpha g_y(x, y), 0, 1),$$

where g_x and g_y are Gaussian-filtered random displacement fields and α is the configured warp scale.

The final unnormalized bathymetry is assembled as

$$\tilde{b}(x, y) = b_{\text{base}}(x, y) + b_{\text{family}}(x_w, y_w) + \sum_i G_i(x, y) + \sum_j R_j(x, y) + N(x, y),$$

where b_{family} is the dominant terrain-family term, G_i are random Gaussian perturbations, R_j are elongated ridge-like structures, and N is multi-scale smoothed roughness. Trench and canyon samples emphasize elongated negative channels, seamount and island samples emphasize localized positive relief, and continental samples combine broad slopes with shelf-break-like structure. The stochastic Gaussian and ridge terms are sign-biased by family so that small-scale structure reinforces the selected morphology rather than erasing it.

After all components are combined, the field is min-max normalized and clipped:

$$b(x, y) = d_{\min} + \frac{\tilde{b}(x, y) - \min_{x,y} \tilde{b}}{\max_{x,y} \tilde{b} - \min_{x,y} \tilde{b}} (d_{\max} - d_{\min}).$$

This normalization deliberately prioritizes morphological diversity over site-specific amplitude realism. It also keeps the numerical problem bounded, which is useful when generating thousands of trajectories for model training. The procedural terrain construction follows the broader idea of synthetic terrain generation for controlled experiments, but the resulting fields should be interpreted as benchmark bathymetries rather than surveyed seafloor products [20, 24].

3.3. Initial-displacement sources

The source generator produces a raw stored source template $q(x, y)$, saved as `source_field`, on the same grid as the bathymetry. The default source family set is

$$\{\text{gaussian}, \text{multi-gauss}, \text{dipole}, \text{fault}, \text{rough}, \text{okada-like}\}.$$

Gaussian and multi-Gaussian sources provide localized smooth uplift patterns. Dipole sources create paired uplift and subsidence lobes. Fault sources create elongated rotated uplift structures. Rough sources are generated from smoothed random fields and broaden the distribution of source geometries. The `okada-like` family samples simplified fault parameters such as length, width, slip, angle, dip, and source depth, and constructs an anisotropic deformation pattern inspired by elastic half-space source models [21]. It is used only as a synthetic source family, not as a full geophysical rupture model.

In the default forward benchmark, per-sample source normalization is disabled. Thus each family preserves its sampled amplitude semantics instead of being forced into a common height interval. A separate independent strength factor

$$s \sim \mathcal{U}(0.45, 0.85)$$

is then applied during scenario construction to form the initial perturbation

$$\eta_0(x, y) = s q(x, y).$$

The initial water depth passed to the shallow-water solvers is

$$h_0(x, y) = \max\{h_{\text{rest}}(x, y) + \eta_0(x, y), 0\},$$

and the horizontal momenta are initialized to zero. Therefore, each rollout starts from a displaced but initially motionless free surface over a synthetic seabed.

An auxiliary Boussinesq diagnostic configuration uses smoother bathymetries, lower source amplitudes, and a smaller source-strength range for propagation checks. The reported Boussinesq dataset uses the same bathymetry and source scenario definitions as the shallow-water solvers, so sample IDs stay aligned across targets.

3.4. Reference rollouts and solver organization

The primary reference labels are generated with finite-volume shallow-water solvers. The hydrostatic-reconstruction solver is used as the main reference because it enforces nonnegative depth and is designed to preserve lake-at-rest balance over non-flat bathymetry [1]. The MUSCL-HR solver adds second-order reconstruction with hydrostatic reconstruction, giving a sharper comparison reference on the same scenarios. Both shallow-water references use Rusanov-type numerical fluxes, open boundary conditions, and a sponge layer near the boundary in the default configuration [16, 28, 5].

For each shared scenario, enabled solvers write outputs into solver-specific sample folders, for example `data/raw/hydrostatic/samples` and `data/raw/muscl_hr/samples`. Each folder stores the full solver state trajectory, the reconstructed free-surface trajectory, timestamps, time-step history, and metadata. For the shallow-water solvers, the stored state contains depth and horizontal momentum components, while the free surface is reconstructed as

$$\eta(x, y, t) = h(x, y, t) + b(x, y).$$

For the Boussinesq solver, the first state channel is treated directly as the free-surface field. Scenario-level manifests record the shared scenario definition, while per-solver manifests record the corresponding numerical labels. This layout allows single-solver training, per-solver comparisons, and multi-reference preprocessing without resampling the underlying bathymetry-source pairs.

Quality checks are applied during generation before a sample is accepted. The default policy rejects non-finite states, depth violations below the configured tolerance, excessive free-surface amplitude, and excessive velocity. These checks are intended to prevent unstable or physically unusable rollouts from silently entering the supervised-learning labels.

3.5. Preprocessing and learning targets

Preprocessing converts the raw rollout archives into model-ready arrays. The preprocessing code supports a default fixed-seed 70%/15%/15% scenario split of a single pool, but for the experiments in this paper we deliberately choose a stronger design and generate the training, validation, and test pools as independent runs. The training pool uses 10,000 scenarios, the validation pool 1,000, and the test pool 2,500. The validation and test pools are generated with separate fixed dataset seeds, so their per-sample random seeds differ even when the sample indices start again from one. This guarantees that no scenario is shared between the pools by construction, rather than relying on a random partition. The same scenario design is reused across solver targets, so the hydrostatic, MUSCL-HR and Boussinesq datasets can be compared without changing the underlying bathymetry-source pairs.

The raw sample stores the source template q , the scalar `source_strength` s , the scaled perturbation `eta0`, the `initial_depth` h_0 , and `free_surface0` separately. The default model input channels are bathymetry, the raw source template q , and initial water depth h_0 ; the scaled perturbation is not used as the default source channel. The initial free surface can also be exported as an ablation channel, and a scalar solver-identity channel is available for explicit multi-reference conditioning experiments. The default target variable is the free-surface elevation η . With the default rollout settings, preprocessing discards the stored initial frame from the prediction target and keeps the next 50 saved frames as a multi-step forecast target.

Normalization statistics are computed from the training split only and then reused unchanged for validation and test samples. In the reported runs, bathymetry and source input channels are standardized with training-split statistics, trajectory targets are standardized for training and denormalized for reported benchmark-scale metrics, and the `initial_depth` input remains in its stored scale under the shared preprocessing configuration. Processed evaluation archives additionally retain sample identifiers, bathymetry type, source type, solver name, source strength, and normalization information. These metadata are used by the evaluation scripts to report aggregate accuracy as well as grouped metrics under distribution-shift suites.

3.6. Distribution-shift and resolution-transfer datasets

The same metadata-first design supports controlled generalization tests. For source-family shift, bathymetry-family shift, and source-strength shift, the evaluation pipeline builds filtered suites from the processed test archives using metadata such as `source_type`, `bathymetry_type`, and `source_strength`. In the current configuration, these suites include the `multi-gauss` subset, the `trench` subset, and the high source-strength tail. These are difficult-subgroup diagnostics because the filtered families are still present in the training distribution.

We also build strict held-out-family datasets for four stress tests: `trench` and `continental` bathymetry holdouts, and `rough` and `okada-like` source holdouts. For these datasets the held-out family is removed before preprocessing and normalization statistics are refit from the filtered training split only. The held-out test subset is then evaluated separately from the in-distribution remainder of the same test pool, which keeps the scenario accounting explicit and avoids confusing metadata filtering with truly unseen-family generalization.

For cross-resolution transfer, we additionally generate matched native 32×32 , 64×64 , and 128×128 auxiliary pools of 1,000 total scenarios per grid, split into 700 training, 150 validation, and 150 test scenarios. These pools keep the same procedural generation logic while changing the grid spacing, the sponge width, and, for the finest grid, the target CFL value. The shared-from-64 preprocessing mode can force the lower- and higher-resolution datasets to use normalization statistics from the 64×64 training distribution. This gives a stricter transfer setting: the model is evaluated on a changed grid resolution without recomputing the feature scaling from the target resolution.

4. Problem formulation and evaluation protocol

This section defines the supervised forward-emulation problem and the evaluation protocol used throughout the benchmark. The data-generation details are described in Section 3; the model architectures and optimization choices are described separately in Sections 5 and 6. Here, the emphasis is on the mathematical object being learned, the numerical references that define the labels, and the metrics used to decide whether a surrogate is accurate, fast, robust, and scientifically interpretable.

4.1. Forward emulation task

Let $\Omega = [0, 1]^2$ denote the normalized two-dimensional computational domain and let $[H, W]$ be the grid resolution. Each scenario contains a bathymetry field $b \in \mathbb{R}^{H \times W}$, a raw stored source template $q \in \mathbb{R}^{H \times W}$, and an initial water-depth field $h_0 \in \mathbb{R}^{H \times W}$. The bathymetry convention is

$$b(x, y) < 0 \quad \text{for submerged cells,}$$

so the still-water depth used by the data generator is

$$h_{\text{rest}}(x, y) = \max(-b(x, y) + \eta_{\text{sea}}, 0),$$

where η_{sea} is the configured sea-level offset. The sampled `source_strength` multiplier s converts the raw source template into the scaled initial free-surface perturbation

$$\eta_0(x, y) = s q(x, y),$$

and the shallow-water initial depth is clipped to remain non-negative,

$$h_0(x, y) = \max(h_{\text{rest}}(x, y) + \eta_0(x, y), 0).$$

The stored initial free surface is therefore $h_0 + b$, which equals η_0 away from the clipping region.

The default supervised input tensor stacks the active spatial channels

$$X = [b, q, h_0] \in \mathbb{R}^{C \times H \times W},$$

with $C = 3$ for the main single-solver experiments. Thus the default source input is the raw template q , not the scaled perturbation η_0 ; the raw archives store `eta0` and `free_surface0` separately for

diagnostics and optional channels. The preprocessing code also supports optional initial-surface and solver-identity channels, but these are not part of the default single-reference FNO configuration. For a numerical reference solver r , the target is a future free-surface trajectory

$$Y_r = [\eta_r(t_1), \eta_r(t_2), \dots, \eta_r(t_T)] \in \mathbb{R}^{T \times H \times W}.$$

The initially stored frame is not used as a prediction target. With the default rollout settings the solver is advanced for 250 updates and stores states with solver-step indices $0, 5, 10, \dots, 250$, giving 51 saved frames per rollout. For the adaptive-CFL shallow-water references, these saved frames are indexed rollout snapshots with recorded timestamps, not values interpolated onto a shared physical-time grid. Preprocessing then discards the initial frame and keeps $T = 50$ future frames as the multi-step forecast target.

A surrogate model f_θ is trained to approximate the solver-conditioned operator

$$f_\theta : X \mapsto \hat{Y}_r \approx Y_r$$

The labels are produced by three numerical references on the same scenario design: a hydrostatic-reconstruction finite-volume solver, a MUSCL-HR finite-volume solver, and a Boussinesq-type weakly dispersive solver. Hydrostatic reconstruction is the primary reference used to anchor the benchmark, while MUSCL-HR and Boussinesq are comparison references with different numerical assumptions. The Boussinesq solver evolves the free surface directly, so its target η is read from the surface state rather than reconstructed from depth; the supervised input, however, uses the same $C = 3$ channel stack in the current pipeline.

4.2. Numerical reference formulation

The shallow-water references evolve the conservative state

$$U = [h, hu, hv]^\top,$$

where h is the water depth and (hu, hv) are the depth-integrated momenta. The free surface is reconstructed as

$$\eta = h + b.$$

Ignoring the boundary terms, the two-dimensional shallow-water equations with bathymetry source terms can be written as

$$\partial_t U + \partial_x F(U) + \partial_y G(U) = S(U, b),$$

with fluxes

$$F(U) = \begin{bmatrix} hu \\ \frac{(hu)^2}{h} + \frac{1}{2}gh^2 \\ \frac{hu hv}{h} \end{bmatrix}, \quad G(U) = \begin{bmatrix} hv \\ \frac{hu hv}{h} \\ \frac{(hv)^2}{h} + \frac{1}{2}gh^2 \end{bmatrix},$$

and the bathymetry source term

$$S(U, b) = \begin{bmatrix} 0 \\ -gh \partial_x b \\ -gh \partial_y b \end{bmatrix}.$$

In the implementation the divisions by h use dry-cell protection and momentum clipping, to avoid numerical overflow near the nearly dry cells.

Both shallow-water references use finite-volume updates of the form

$$U_{i,j}^{n+1} = U_{i,j}^n - \frac{\Delta t}{\Delta x} (\mathcal{F}_{i+1/2,j} - \mathcal{F}_{i-1/2,j}) - \frac{\Delta t}{\Delta y} (\mathcal{G}_{i,j+1/2} - \mathcal{G}_{i,j-1/2}) + \Delta t S_{i,j},$$

where $\mathcal{F}$ and $\mathcal{G}$ are the numerical interface fluxes. The base interface flux is a Rusanov flux,

$$\mathcal{F}(U_L, U_R) = \frac{1}{2} (F(U_L) + F(U_R)) - \frac{1}{2} \alpha_x (U_R - U_L),$$

with local wave-speed estimate

$$\alpha_x = \max(|u_L| + \sqrt{gh_L}, |u_R| + \sqrt{gh_R}),$$

and an equivalent y -direction speed using v . This choice is diffusive compared with sharper Riemann solvers, but it is robust and suitable for controlled dataset generation [28, 29].

Because the scheme is explicit, the time step is not free: an explicit finite-volume update is only stable when the information does not travel more than one cell per step, which is the Courant–Friedrichs–Lewy (CFL) condition. We therefore do not fix Δt in advance but choose it adaptively at every step from the current state, using the same characteristic speed $|U| + \sqrt{gh}$ that appears in the Rusanov flux,

$$\Delta t = \text{CFL} \cdot \left[\max \left(\frac{\max(|u| + \sqrt{gh})}{\Delta x}, \frac{\max(|v| + \sqrt{gh})}{\Delta y} \right) \right]^{-1},$$

where the maxima are taken over the whole grid and the target CFL number is $\text{CFL} = 0.45$ in the default configuration. Adapting Δt this way keeps the rollout stable when a large initial displacement temporarily increases the wave speed, while still taking the largest safe step elsewhere; this matters for a benchmark, because an over-large fixed step would let unstable samples enter the training labels.

The hydrostatic-reconstruction solver modifies the left and right water depths at each face before computing the Rusanov flux. For an interface with bed elevations b_L and b_R , it sets

$$z^* = \max(b_L, b_R), \quad h_L^* = \max(0, h_L + b_L - z^*), \quad h_R^* = \max(0, h_R + b_R - z^*).$$

These reconstructed states preserve nonnegative water depth and help to balance the flux-gradient and bed-slope terms for stationary water over uneven bathymetry [1]. In this well-balanced formulation the bathymetry source S is absorbed into the reconstructed face fluxes rather than added as a separate term, so the $\Delta t S_{i,j}$ contribution above is schematic for the hydrostatic solver; the MUSCL-HR solver instead applies an explicit, centered discretization of the same source after its slope-limited reconstruction. The MUSCL-HR solver uses the same hydrostatic face idea but reconstructs h , η , u and v with minmod-limited slopes before face update, followed by a two-state Heun update. This gives a less diffusive second-order comparison reference while retaining the lake-at-rest safeguards needed for tsunami-like bathymetry [5, 9].

The third numerical reference is a Boussinesq-type comparison solver. We include it because the shallow-water equations above are non-dispersive: all long waves travel according to the local depth-dependent speed, but the model does not represent the weak frequency dispersion that can matter

for shorter components of a tsunami-like disturbance. The implementation is not a full nonlinear coastal Boussinesq model. It is an elevation-only, linear, weakly dispersive approximation used as an additional reference beside Hydrostatic and MUSCL-HR.

In the variable-depth mode used here, the unknown is the free-surface elevation η and its time derivative η_t . If the dispersive correction is removed, the model reduces to the familiar linear long-wave surface equation

$$\eta_{tt} = g \nabla \cdot (H \nabla \eta),$$

where H is the still-water depth. The implemented Boussinesq form modifies this by solving for the acceleration $a_\eta = \eta_{tt}$ through a Helmholtz-type linear system,

$$(I - \alpha_B \nabla \cdot (H^2 \nabla)) a_\eta = g \nabla \cdot (H \nabla \eta),$$

where

$$H(x, y) = \max((-b(x, y) + \eta_{\text{sea}}) s_H, H_{\min})$$

is the effective depth used inside this solver. The factor s_H is an implementation parameter used only for auxiliary diagnostics; in the shared dataset reported here, we use $s_H = 1$. The lower bound $H_{\min}$ prevents zero or near-zero depth from making the operator singular or badly conditioned. In the main configuration $H_{\min} = 10^{-4}$, $\alpha_B = 1/3$, and the same normalized bathymetry convention as the shallow-water solvers is used.

The coefficient α_B controls the strength of the weakly dispersive correction. When $\alpha_B = 0$, the left hand side becomes just a_η , so the update is just the non-dispersive linear long-wave equation written above. A positive α_B makes the acceleration depend on a spatial solve involving $H^2 \nabla a_\eta$, which is the part that gives this reference its weakly dispersive character. We set $\alpha_B = 1/3$, matching the solver default, and treat it as a fixed model choice rather than a fitted physical parameter. The linear system is applied matrix-free and solved with conjugate gradients, so the implementation never forms a dense matrix; the rollout is then advanced with a velocity-Verlet step, which updates η and η_t using accelerations at the start and end of the step. This gives the benchmark a weakly dispersive comparison reference beside the two shallow-water references, while still keeping Hydrostatic as the main solver and without treating any numerical reference as real-ocean truth.

4.3. Reference-solver health and consistency checks

As health and consistency checks for the reference solvers used for dataset generation, we run a small series of numerical tests before generation. These checks screen for basic implementation failures and unstable labels; they are not substitutes for grid/time-step convergence studies, independent solver benchmarks, or event-scale validation against observations.

Lake at rest. A well-balanced shallow-water solver should preserve a stationary free surface over non-uniform bathymetry. We therefore initialize a fully submerged, spatially varying bathymetry with zero surface elevation, $\eta_0(x, y) = 0$, and zero momentum, $hu(x, y) = hv(x, y) = 0$. Since $\eta = h + b$, this corresponds to an initial water depth $h_0(x, y) = -b(x, y)$. Both the hydrostatic and MUSCL-HR solver are advanced for many adaptive-CFL time steps with the sponge damping disabled. We consider the lake-at-rest condition preserved when the maximum free-surface drift

stays below 10^{-8} and the maximum absolute momentum stays below 10^{-7} . This property matters over long rollouts, because a solver that slowly drifts from rest would inject a spurious signal into every training label.

Dynamic finite-state behavior. Lake at rest only checks the trivial equilibrium, so we additionally run dynamic rollouts from non-trivial initial displacements and confirm that the state remains finite and bounded over the full horizon. This guards against late time blow-up or oscillations that look stable for a few steps and then produce non-finite values near the dry cells.

Quality status. During generation each rollout is screened by the quality policy, which records whether the trajectories contain non-finite values, depth below tolerance, or excessive surface amplitude and velocity. For the Boussinesq solver, the same health record also checks the free-surface-to-depth ratio and whether the conjugate-gradient acceleration solve converged. The outcome is written into the per-sample metadata as a quality status. This status also makes accepted samples explicitly identifiable during dataset construction, avoiding unnecessary recomputation.

4.4. Evaluation metrics

For field accuracy, we report the mean absolute error (MAE), root-mean-squared error, the relative L_2 error, and maximum absolute error over the predicted trajectory. Writing y for the reference frames and $\hat{y}$ for the prediction, and letting k index all elements of the $T \times H \times W$ trajectory tensor with N entries in total,

$$\begin{aligned} \text{MAE} &= \frac{1}{N} \sum_k |\hat{y}_k - y_k|, & \text{RMSE} &= \sqrt{\frac{1}{N} \sum_k (\hat{y}_k - y_k)^2}, \\ \text{rel } L_2 &= \frac{\|\hat{y} - y\|_2}{\|y\|_2}, & \text{max-err} &= \max_k |\hat{y}_k - y_k|. \end{aligned}$$

Unless otherwise stated, MAE and RMSE are computed after reversing the target normalization, so that those errors are reported in denormalized benchmark-scale surface-elevation units. The relative L_2 error is dimensionless. The max-err values in the main tables are tracked with the same aggregation logic but are reported in normalized target units unless a caption explicitly marks them as denormalized or physical; this maximum remains a useful proxy for local worst-case behavior and for time-stepping stability.

When we aggregate these metrics over a whole evaluation split, we accumulate the underlying sums across every sample first and only reduce at the end, instead of averaging the per-batch numbers. Concretely we keep a running total of $\sum_k |\hat{y}_k - y_k|$, $\sum_k (\hat{y}_k - y_k)^2$, and $\sum_k y_k^2$ over the full split, and track a single running maximum for max-err. This matters because the relative L_2 is a ratio of norms and the max-err is a maximum, so averaging them batch-by-batch would not give the true split-level value: a batch-averaged max-err in particular only reports the mean of the per-batch maxima and therefore understates the real worst case. Computing them from the accumulated sums means the reported rel L_2 and max-err are the genuine dataset-level quantities, and they no longer depend on the batch size used at evaluation time.

For distribution-shift evaluation the metrics are computed per sample first and then grouped by metadata, using the source type, the bathymetry type, the source-strength range, or the reference solver. Computing per-sample errors before grouping avoids letting a few large scenarios dominate a suite average. Note that this is deliberately a different aggregation from the split-level accumulation described above: here the relative L_2 and max-err of a group are the mean of the per-sample values, not a single global ratio or global maximum, so the two kinds of table should not be compared cell-for-cell. In addition to the field metrics above, we report a set of frequency-band and conserved-quantity diagnostics in the appendix, following the metric design of PDEBench [27]; these are useful for separating low and high-wavenumber error, which is relevant when the surrogate is a spectral operator.

4.5. Generalization and resolution-transfer protocol

The metadata-filtered suites whose composition is described in Section 3 are evaluated under a fixed protocol. For each suite we score the trained checkpoint on the held-out samples using the same per-sample-then-grouped metrics as above, so that an in-distribution number and an out-of-distribution number are always produced by the identical pipeline and differ only in the scenarios that are included. We distinguish two cases that are easy to conflate. A *difficult-subgroup* test keeps a family in the training distribution but reports the error on a metadata-selected subset of the test pool, while a *held-out-family* test removes that family from the training split entirely before preprocessing and only then evaluates it. We label each suite by which of the two it is, because only the second is evidence of generalization to an unseen family.

For resolution transfer we keep two settings separate. The *proxy* setting resizes the 64×64 test fields to a target grid and evaluates the same checkpoint on the resized inputs; it is a cheap diagnostic and does not involve any new simulation, so it is not a like-for-like simulation-resolution result. The *native* setting instead evaluates the checkpoint on data that were independently generated and simulated at 32×32 and 128×128 , which is the stronger test because the wavefield at each resolution is physically resolved rather than interpolated. Cross-resolution comparison is only meaningful when the target normalization is controlled, so we adopt a strict policy: a native-resolution suite is accepted only when its target normalization statistics match the reference statistics of the training distribution, and a transfer claim that mixes incompatible statistics is rejected rather than silently reported. This prevents a change of feature scaling from being mistaken for genuine resolution generalization.

4.6. Runtime benchmarking

The numerical reference solvers are timed using their CPU NumPy rollout implementations, excluding the bathymetry and source generation, so the denominator reflects only the forward integration. The neural surrogate is timed on CPU and, when available, the GPU. For every measurement we record the device, the batch size, the numerical precision label, and the TF32 back-end state, and on GPU we synchronize the device before and after timing so that asynchronous kernel launches are not counted as zero. Each configuration uses a short warmup and is then repeated several times, and we report the per-sample time and throughput in samples per second.

The speedup is computed as

$$\text{speedup} = \frac{\text{reference-solver rollout time per sample}}{\text{surrogate inference time per sample}}.$$

We stress that these speedups are implementation-level comparisons under the reported hardware, and not universal comparisons against optimized production-grade tsunami solvers. The CPU NumPy references are reproducible and easy to audit, but they are not tuned for speed, so the numbers should be read as a relative indication and not as an absolute performance claim.

4.7. Cross-solver and emulator-superiority analysis

Because the training solver is not itself the ground truth, we separate two different quantities. The first is the raw solver-vs-solver gap, that is the denormalized benchmark-scale difference between two numerical references evaluated on the same shared scenarios at matching saved-frame indices. The second is the surrogate error of a model trained on one reference and compared against another. Keeping these apart is important, since a surrogate can look poor only because the two reference solvers themselves disagree. These are released-label, same-scenario, same-saved-index diagnostics, not strict common-physical-time solver-fidelity measurements.

Following the neural emulator-superiority idea [12], we also report a ratio that compares a surrogate trained on reference A against a second reference B .

$$\rho = \frac{\text{Err}(f_{\theta}^A, \text{ref } B)}{\text{Err}(\text{ref } A, \text{ref } B)},$$

where the error is a denormalized benchmark-scale metric evaluated on the same scenarios with separate denormalization for the prediction and target. A value $\rho < 1$ would mean that the emulator trained on A is in fact closer to B than its own training solver A is. We only report this ratio when the prediction and target denormalization, and the input normalization statistics, are compatible, because otherwise the scale comparison is not meaningful. The hydrostatic, MUSCL-HR, and Boussinesq references are generated on the same scenario IDs, so raw solver-vs-solver field gaps can be computed for all pairs. The emulator-superiority ratio is treated as a main result for the hydrostatic/MUSCL-HR pair because those are the two shallow-water references; Boussinesq is reported as a weakly dispersive comparison reference for the same benchmark rather than as physical truth or an automatic upgrade.

4.8. Uncertainty quantification

Point accuracy alone does not tell a user when a surrogate is likely to be wrong, which matters for any hazard-oriented application. We therefore estimate predictive uncertainty with a deep ensemble of M independently trained models that differ only in their random seed [14]. For an input X the members produce predictions $\{\hat{Y}^{(m)}\}_{m=1}^M$, and we form the ensemble mean and the per element predictive variance

$$\bar{Y} = \frac{1}{M} \sum_{m=1}^M \hat{Y}^{(m)}, \quad \sigma^2 = \frac{1}{M} \sum_{m=1}^M \left(\hat{Y}^{(m)} - \bar{Y} \right)^2$$

where $\bar{Y}$ is used as a prediction and σ as the uncertainty. A single member gives a degenerate zero variance, so we require $M \geq 2$ and use $M = 7$ in the reported experiment.

We assess the quality of this uncertainty in two ways. First, we measure whether the uncertainty tracks the actual error through the Pearson correlation between the per-element predictive standard deviation σ and the absolute error $|\bar{Y} - Y|$; a useful uncertainty should be larger where the error is larger. Second, we check calibration by treating each prediction as Gaussian with mean $\bar{Y}$ and standard deviation σ and measuring the empirical coverage of central prediction intervals at several nominal levels, together with the Gaussian negative log-likelihood. A well-calibrated model should have an empirical coverage close to the nominal level, for example about 90% of targets inside the nominal 90% interval. We additionally report these uncertainty metrics on the distribution-shift suites of Section 4.4, since a trustworthy uncertainty should grow under distribution shift rather than stay falsely confident. We interpret these as reliability diagnostics and do not claim formally calibrated risk.

5. Surrogate models

All surrogates share the same interface fixed in Section 4.1: they map the stacked input field $X \in \mathbb{R}^{C \times H \times W}$ to the full forecast trajectory $\hat{Y} \in \mathbb{R}^{T \times H \times W}$, with $C = 3$ and $T = 50$ in the main experiments. The forecast horizon is therefore predicted as T output channels, so that a whole rollout is produced in a single forward pass. We keep this interface identical across all models so that differences in the results reflect the architecture and not a different problem setup. The Fourier Neural Operator is the main model; the others are baselines that span a local and an encoder–decoder inductive bias.

5.1. Fourier Neural Operator

The Fourier Neural Operator (FNO) is our primary model because it learns a mapping between discretized functions rather than a fixed pixel-to-pixel function, and its spectral layer gives it a global receptive field at every depth [18]. This suits tsunami-like propagation, where a disturbance influences distant parts of the domain within the forecast horizon and a purely local model would need many layers to reach the same range.

The overall architecture is shown in Figure 2. The input is first lifted to a width- d feature map by a 1×1 convolution. Following common practice, we also concatenate two normalized coordinate channels before lifting, so the operator has access to absolute position. The lifted features are then processed by a stack of L Fourier blocks. Each block combines a spectral convolution with an ordinary 1×1 convolution and a GELU nonlinearity,

$$z \mapsto \text{GELU}(\mathcal{K}(z) + Wz),$$

where W is the pointwise (local) path and $\mathcal{K}$ is the spectral (global) path. The spectral path transforms the features with a two-dimensional real Fast-Fourier-Transform (FFT), keeps only the lowest $m_1 \times m_2$ Fourier modes, multiplies them by a learnable complex weight tensor, and maps back with an inverse FFT. Truncating to the low modes is both a regularizer and the reason the layer is resolution-flexible: because the trained mode count is fixed, the full FFT grid is rebuilt from

the current input size, so the same trained weights can be applied at a different grid resolution, which is what we exploit in the cross-resolution experiments. After the Fourier blocks, a two-layer pointwise head with GELU in between projects the width- d features to the T output frames. To reduce warp-around artifacts from a periodic FFT on a non-periodic domain, the features are slightly zero-padded before the blocks and cropped back afterwards.

In the main configuration we use $L = 6$ Fourier blocks, width $d = 48$, and $m_1 = m_2 = 14$ retained modes in each spatial direction, with a padding of 6 cells. The same architecture is trained separately on the hydrostatic, MUSCL-HR, and Boussinesq labels; only the output channel count would change if the forecast horizon were changed. We also train two hydrostatic mode-count ablations, FNO-modes8 and FNO-modes20, that keep the rest of the configuration fixed while changing the retained Fourier modes to 8 and 20 per axis.

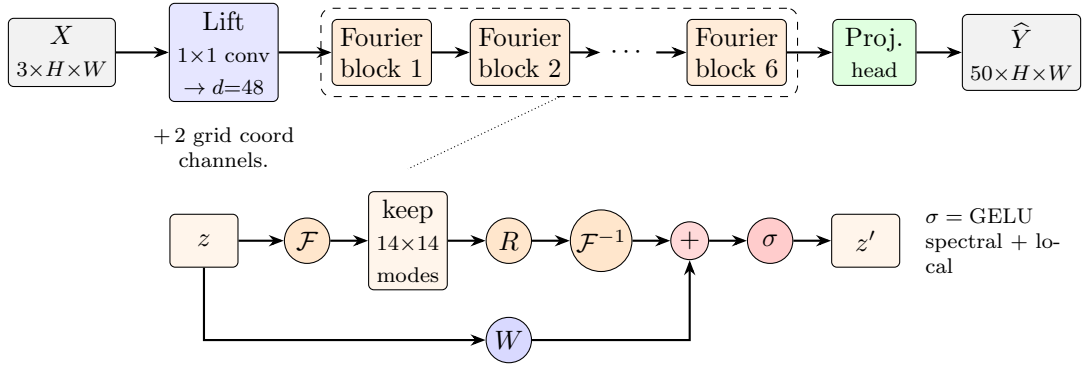


Figure 2: FNO architecture used in this work. **Top:** the input field X (bathymetry, raw source template, initial depth; $C=3$) plus two grid-coordinate channels is lifted to width $d=48$, passed through $L=6$ Fourier blocks, and projected to the $T=50$ forecast frames. **Bottom:** one Fourier block. The spectral path applies a real FFT, keeps the lowest 14×14 modes, multiplies them by a learnable complex weight R , and inverts the transform; the local path W is a 1×1 convolution. Their sum passes through a GELU. Truncating to the low modes is what lets the same weights run at a different grid resolution.

5.2. Operator variants

The final comparison set includes three additional neural-operator variants under the same direct single-pass interface. The **FFNO** replaces the dense two-dimensional spectral mixing with a factorized Fourier structure, reducing the parameter coupling across axes while preserving global spectral propagation [30]. The **U-FNO** combines Fourier blocks with a U-Net-like multiscale path [32], and the **WNO** replaces the Fourier basis with a wavelet-style multiresolution operator [31]. These are used as established architecture baselines rather than as new methods proposed here, so they share the same input channels, target frames, normalization, loss, and evaluation metrics as the main FNO wherever possible.

5.3. Windowed rollout variants

The direct models predict all 50 target frames from the static input stack alone. For the long-horizon drift diagnostic, we also train Window-FNO-5 and Window-FFNO-5 on the hydrostatic reference. These models forecast in blocks of $K = 5$ frames from a dynamic stack that includes bathymetry, the raw source template, and the two most recent surface-elevation frames. At evaluation time the rollout is seeded with the first target frame and then advanced autoregressively, with metrics computed over the remaining 49 predicted target frames, so the windowed results are reported as first-target-frame-seeded diagnostics rather than as like-for-like replacements for the direct 50-frame single-pass task.

5.4. Convolutional baselines

To check whether the spectral, global structure of the FNO actually helps, we include two purely convolutional baselines that have a local receptive field.

The **CNN** is a small fully convolutional neural network: 3×3 convolutions followed by a GELU activation, then a 1×1 output convolution that produces T output frames at a width- $d = 48$. It has no down-sampling and no global mixing, so its receptive field grows only with depth. It therefore represents the simplest reasonable local predictor and is useful as a lower comparison anchor.

The **U-Net** adds a single encoder-decoder with a skip connection [23]. An input double convolution block is followed by average pooling with down-sampling and a second double convolution at double the width; the coarse features are then bilinearly upsampled, concatenated with a high-resolution skip feature, and decoded by a final double convolution and 1×1 output convolution at a base width of 32. The pooling gives it a larger effective receptive field compared to the plain **CNN** at a modest cost, which is why it is a stronger convolution comparison.

5.5. Ensemble uncertainty

For uncertainty, we do not change the base architecture; we instead train a deep ensemble of independently seeded FNO models and aggregate their predictions, as defined in Section 4.8. The ensemble mean is used as the point prediction and the per-element spread across members as the uncertainty, which keeps the uncertainty mechanism simple and architecture-agnostic. Because a single member gives a degenerate zero variance, the ensemble path requires at least two members, and we use seven in the reported experiments.

6. Training setup

6.1. Data splits and normalization

All models are trained on the processed datasets described in Section 3. As explained there, the training, validation, and test scenarios come from separately generated pools with separate fixed

dataset seeds rather than from a random partition of the pool, which guarantees that no scenario is shared between them. The main experiment uses 10,000 training scenarios, 1,000 validation scenarios, and 2,500 test scenarios per solver target, with the same scenario design reused across the hydrostatic, MUSCL-HR, and Boussinesq labels so the targets are directly comparable.

Normalization statistics are estimated on the training split only and then applied unchanged to the validation and test splits, so that no test information leaks into the feature scaling. Bathymetry and source input channels are standardized with training-split statistics. Trajectory targets are standardized for training and denormalized at evaluation time for the reported benchmark-scale metrics. The `initial_depth` channel follows the shared preprocessing configuration and remains in stored scale for the reported runs.

6.2. Optimization

All models are trained with the AdamW optimizer at an initial learning rate of 5×10^{-4} and a weight decay of 10^{-6} . The learning rate follows a cosine-annealing schedule that decays to a floor of 5×10^{-5} over the training budget, and the gradients are clipped to a maximum norm of 1.0 for stability. We use a batch size of 64.

Because the target is a 50-frame trajectory, we train with a horizon-weighted mean-squared error that increases the weight of later frames,

$$\mathcal{L} = \frac{\sum_{t=1}^T w_t \text{MSE}_t}{\sum_{t=1}^T w_t}, \quad w_t = w_{\min} + (w_{\max} - w_{\min}) \left(\frac{t-1}{T-1} \right)^p,$$

where MSE_t is the mean-squared error of frame t and we use $w_{\min} = 0.8$, $w_{\max} = 2.0$, and $p = 1.7$. Without this weighting, the optimizer tends to fit the easy early frames and leave the later, lower-amplitude part of the rollout blurry, which is exactly the regime that matters for a forecast horizon. The relative L_2 error on the validation split is used both as the checkpoint-selection metric, keeping the best epoch, and as the early-stopping criterion with a patience of 20 epochs. One detail worth stating: this selection metric is the relative L_2 computed on the *normalized* validation targets, whereas the accuracy tables we report later are in denormalized benchmark-scale units after the normalization is reversed. Because standardization subtracts the channel mean, the normalized and denormalized relative L_2 have different denominators and are not the same number, so the best epoch is chosen on the normalized criterion and then scored on the denormalized metrics rather than being selected directly on the denormalized error. To keep the comparison fair, the same-protocol direct models share a common upper training budget: each model is allowed up to 256 epochs, the largest budget used by any model, and the same optimizer, schedule, loss, and stopping rule are applied to all of them unless a model-specific configuration explicitly changes only the architecture. Early stopping then halts each model on its own, when its validation error stops improving, so a faster-converging baseline is not penalized and a slower model is not cut short. In practice the budget is an upper bound rather than a fixed number of epochs, and most models stop well before reaching it.

6.3. Experiment configurations

Each model–target pair is defined by a single configuration file. The primary configurations train the Fourier Neural Operator on the hydrostatic, MUSCL-HR, and Boussinesq labels, and the hydrostatic comparison set additionally includes FFNO, FNO-modes8, FNO-modes20, U-FNO, WNO, CNN, and U-Net models under the matched protocol above. Hydrostatic is the main reference target, while MUSCL-HR and Boussinesq are the main comparison targets; Boussinesq is included because it gives a weakly dispersive baseline on the same scenarios, not because it is treated as a validated truth model.

The windowed configurations use the same hydrostatic data but change the supervised task from a direct 50-frame forecast to a $K = 5$ frame block forecast conditioned on recent surface-elevation frames. The reported Window-FNO-5 and Window-FFNO-5 rollouts are first-target-frame-seeded diagnostics at evaluation time and scored on the remaining 49 predicted target frames, so they are used as drift diagnostics and are not treated as like-for-like with the direct single-pass prediction problem.

For the uncertainty experiments we train a deep ensemble of seven members that differ only in their random seed, following the procedure of Section 4.8. The ensemble shares the architecture and the training protocol of a single FNO model and only changes the seed, so any spread between members reflects optimization stochasticity rather than a different model design.

6.4. Hardware and runtime setting

Dataset generation was performed on an 8-core AMD EPYC 9B45 CPU server, because the numerical reference solvers are the main computational bottleneck during benchmark construction. In particular, each scenario must be advanced by explicit numerical time integration before it can become a supervised training label. Model training and evaluation were then run on accessible local hardware: an Intel Core i5-13420H CPU and an NVIDIA GeForce RTX 4050 Laptop GPU. We report this setting explicitly because one goal of the benchmark is to make neural-surrogate experimentation feasible for smaller laboratories, student researchers, and local communities without access to large GPU clusters.

The resulting runtime numbers should therefore be interpreted as measurements under a resource-conscious research setup rather than as optimized production benchmarks. They quantify the acceleration of the neural surrogate relative to the reproducible reference implementations used in this study, not relative to highly optimized operational tsunami models or GPU-accelerated numerical solvers. A global random seed of 42 is set for the Python, NumPy, and PyTorch random number generators, the ensemble members use the fixed seed $\{11, 22, 33, 44, 55, 66, 77\}$, and the data splits are fixed by construction as described above. Every run writes its resolved configuration, its full epoch history, and its best and last resolved checkpoints into a per-run directory. The speed-evaluation outputs also record the selected device, Python/PyTorch/CUDA versions, precision label, and TF32 backend state, so each reported runtime measurement can be traced back to the configuration and hardware that produced it.

7. Results

All accuracy numbers are reported on the held-out test split ($N = 2500$ scenarios). Unless noted, MAE and RMSE are in denormalized benchmark-scale surface-elevation units obtained by reversing the target normalization, relative L_2 is dimensionless, and max-err is in normalized target units. Following Section 4.4, the relative L_2 and max-err values are accumulated globally over the whole split rather than averaged per batch. The FNO is trained and evaluated separately against each of the three numerical references; the expanded hydrostatic comparison set includes FFNO, mode-count ablations, U-FNO, WNO, U-Net, and CNN models.

7.1. Reference-solver health checks and dataset quality

All three numerical references use the same scenario design: parallel hydrostatic, MUSCL-HR, and Boussinesq datasets of 13,500 scenarios each, split into 10,000 train, 1,000 validation, and 2,500 test rollouts with fixed, disjoint seeds. Every target is a 50-frame η trajectory on a 64×64 grid with the $C = 3$ input stack of Section 4.1. Under the quality policy of Section 4.3, all 13,500 scenarios in each dataset pass with `quality_status = ok`: no rollout is dropped for non-finite values, sub-tolerance depth, or amplitude violations, and the Boussinesq conjugate-gradient acceleration solve converged on every sample. Source strength spans the same $[0.45, 0.85]$ range, with the six source families and five bathymetry types matched across references and splits, so each surrogate sees the same scenario population. Because scenario IDs are shared, the cross-solver gaps of Section 7.7 use all 2,500 shared test scenarios for each pair, but they are same-scenario, same-saved-index released-label diagnostics rather than interpolated common-physical-time comparisons. The timestamp audit in Appendix A.2 confirms that Hydrostatic and MUSCL-HR frame 50 are nearly time-aligned, while Boussinesq frame 50 is at about half the shallow-water physical time.

Table 1 summarizes the final solver health and consistency checks. The 24-sample lake-at-rest tests preserve roundoff, all 13,500 stored dynamic rollouts per solver are finite, and Boussinesq has zero recorded conjugate-gradient failures. With sponge disabled for the conservation diagnostic, the shallow-water references keep the mass/free-surface integral to about 2.6×10^{-16} . The CFL rows are only a timestep-sensitivity diagnostic at matched benchmark end time: CFL 0.60 remains finite on the subset but shows large final-field differences, so these numbers are not a formal convergence proof. Taken together, these are sanity, sensitivity, and health checks, not substitutes for a convergence study, external solver comparison, observed-event validation, or production-code validation.

Table 1: Reference-solver sanity/health-check summary. Lake-at-rest and conservation checks are rerun diagnostics on denormalized inputs; dynamic finite-state statistics are aggregated over the generated datasets. CFL sensitivity is diagnostic only, not a formal convergence result.

Check	Solver	Samples	Result
Lake at rest max η drift	Hydrostatic	24	8.88×10^{-16}
Lake at rest max η drift	MUSCL-HR	24	5.33×10^{-15}
Lake at rest max η drift	Boussinesq	24	0
Dynamic finite pass rate	Hydrostatic / MUSCL-HR / Boussinesq	13,500 each	1.000
Boussinesq CG failures / max iterations	Boussinesq	13,500	0 / 416
Max mass rel. change	Hydrostatic / MUSCL-HR	24 each	$\approx 2.6 \times 10^{-16}$
CFL final-field sensitivity vs 0.45	Hydrostatic / MUSCL-HR	12 each	diagnostic

7.2. Same-solver surrogate accuracy

Table 2 reports direct single-pass accuracy on the hydrostatic reference, plus the FNO trained against each reference. Among the single-run direct hydrostatic configurations tested, FFNO is strongest, reaching relative $L_2 = 0.168$ with 0.411M parameters, compared with 0.235 and 5.443M for the base FNO (about $13\times$ fewer). The base FNO, U-FNO, and FNO-modes20 are effectively tied around 0.235; FNO-modes8 is slightly worse at 0.241; and WNO, U-Net, and CNN trail at 0.367, 0.475, and 0.607. The ordering supports global operator structure, but the small gap between Fourier-family variants suggests the main gain is not simply from increasing retained modes.

Table 2: Same-solver field accuracy on the held-out test split ($N = 2500$). Parameter counts are in millions. MAE and RMSE are in denormalized benchmark-scale surface-elevation units; relative L_2 is dimensionless; max-err is the global worst-case element in normalized units. The first block shares the hydrostatic reference; the last two rows are the FNO trained and scored against the MUSCL-HR and Boussinesq references on their own test splits.

Model	Reference	Params (M)	MAE	RMSE	rel- L_2	max-err
FFNO	Hydrostatic	0.411	0.00066	0.00163	0.168	18.6
FNO	Hydrostatic	5.443	0.00087	0.00227	0.235	22.9
U-FNO	Hydrostatic	6.938	0.00086	0.00227	0.235	26.3
FNO-modes20	Hydrostatic	11.083	0.00088	0.00228	0.235	24.4
FNO-modes8	Hydrostatic	1.793	0.00084	0.00233	0.241	25.5
WNO	Hydrostatic	2.016	0.00151	0.00355	0.367	23.3
U-Net	Hydrostatic	0.104	0.00187	0.00460	0.475	27.4
CNN	Hydrostatic	0.045	0.00236	0.00588	0.607	29.8
FNO	MUSCL-HR	5.443	0.00119	0.00281	0.263	22.1
FNO	Boussinesq	5.443	0.00137	0.00468	0.210	15.9

Across references, FNO relative L_2 stays between 0.21 and 0.26. It is lowest on the Boussinesq target (0.210), but this ranking is unit-dependent: in denormalized benchmark-scale RMSE, the Boussinesq surrogate is the *least* accurate same-solver FNO run (0.00468 vs 0.00227 hydrostatic and 0.00281 MUSCL-HR). The dimensionless relative L_2 and normalized-unit max-err metrics are affected by the Boussinesq target normalization scale, about $2.3\times$ larger (0.0221 vs 0.0097). We therefore avoid calling any single reference “best” outright: the FNO emulates all three to comparable dimensionless fidelity, while the denormalized benchmark-scale RMSE ranking favors the shallow-water targets. These are same-solver numbers, each surrogate scored against the reference it was trained on, so they measure emulation quality and are kept separate from the raw solver-vs-solver gaps in Section 7.7.

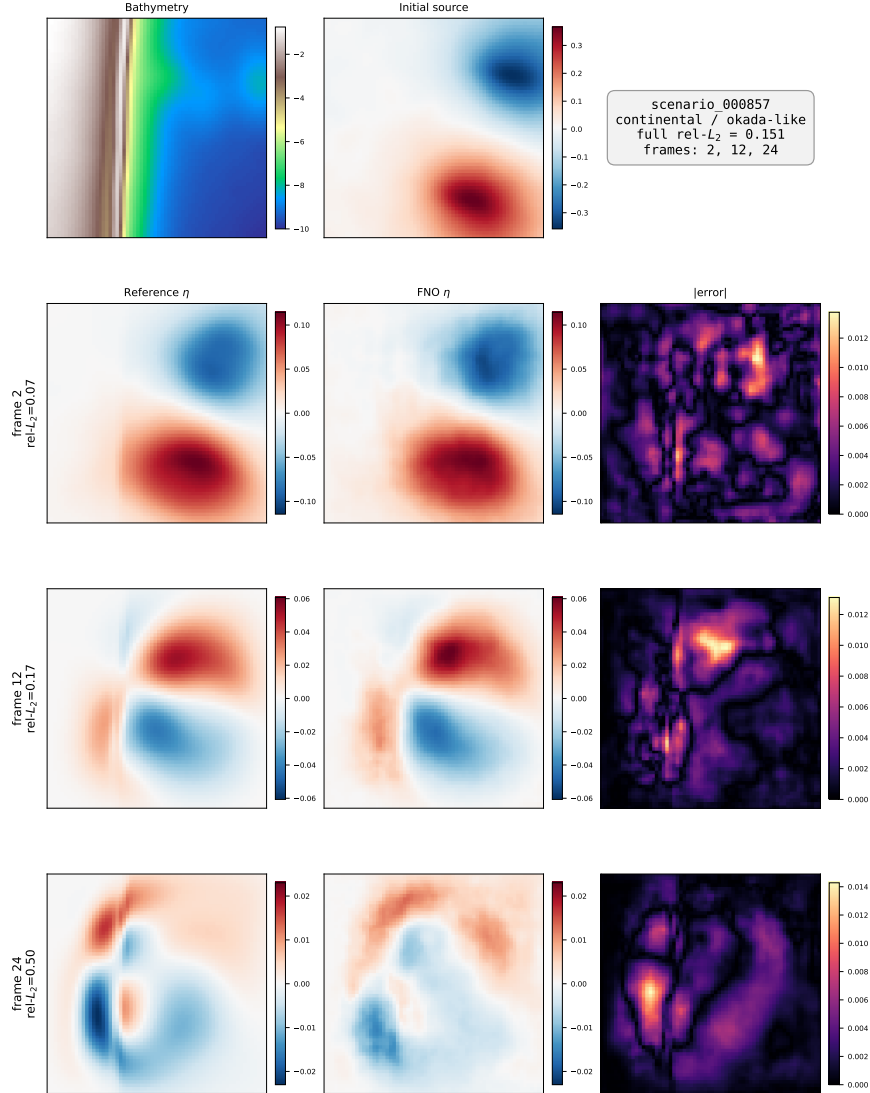


Figure 3: Single-pass FNO rollout on one hydrostatic test scenario. The top row shows denormalized bathymetry, source, and metadata; lower rows show reference η , prediction, and denormalized benchmark-scale absolute error for early, middle, and late frames. Reference and prediction share each frame’s symmetric η scale; the changing scale reflects amplitude decay over the horizon, and late-frame error growth matches Table 3.

Per-frame error growth. The single-pass models predict all 50 frames in one shot, but their accuracy is not uniform over the forecast horizon, as the qualitative rollout in Figure 3 already suggests. Table 3 reports the per-frame relative L_2 at the first, middle, and last frame for the FNO on each reference and for the stronger hydrostatic FFNO. On the hydrostatic target the FNO error rises over the horizon, from 0.214 at the first frame to a peak of 0.355 late in the rollout (around frame 38), then eases slightly to 0.309 at the final frame, so the last frame is somewhat below the mid-to-late peak rather than the maximum. The FFNO lowers the whole curve, from 0.135 at the first frame to 0.273 at the last, but its peak still rises to 0.289, so factorization reduces rather than removes drift. On MUSCL-HR the FNO error grows more steeply and does not recover, from 0.217 to 0.469 at the last frame – the worst late-horizon drift of the three. This is a limitation of a horizon-predicting surrogate: even without an explicit autoregressive feedback loop, late frames depend on longer-range propagation and accumulate more error, the same drift pattern a recurrent rollout would show. The Boussinesq FNO is the clear exception: its per-frame error is essentially flat, 0.207 at the first frame and 0.208 at the last (range 0.205–0.217), showing no temporal drift over the horizon. We attribute this to the smoothness of the weakly dispersive elevation field discussed above rather than to a different model, and we return to the drift-targeting windowed forecaster in Section 7.8.

Table 3: Per-frame relative L_2 at the first frame, the mid-horizon frame, and the last frame of the 50-step rollout ($N = 2500$). Numbers are in normalized units. FFNO reduces hydrostatic drift but does not remove it; the Boussinesq FNO remains nearly flat.

Model	Reference	Frame 1	mid-horizon	frame 50
FFNO	Hydrostatic	0.135	0.264	0.273
FNO	Hydrostatic	0.214	0.325	0.309
FNO	MUSCL-HR	0.217	0.385	0.469
FNO	Boussinesq	0.207	0.206	0.208

Physics diagnostics. Table 4 gives three compact physics-aware diagnostics for the hydrostatic FNO and FFNO. The free-surface integral error measures the spatial-mean η time series, the mass proxy uses the fully wet relation $h = \eta - b$, and the spectral-band rows split trajectory error by Fourier radius. These diagnostics track where the surrogate errors live; they do not imply that either network enforces conservation. The FFNO improves all three summaries, lowering the free-surface integral relative L_2 from 0.190 to 0.139 and the mass-proxy error from 1.20×10^{-4} to 8.77×10^{-5} , while both models remain weakest in the high-wavenumber band.

Table 4: Hydrostatic physics diagnostics for direct single-pass FNO and FFNO on the held-out test split. Integral diagnostics are relative L_2 of spatial-mean time series; spectral columns are relative L_2 by Fourier band.

Model	η integral	mass proxy	spectral low	spectral mid	spectral high
FNO	0.190	1.20×10^{-4}	0.220	0.499	0.687
FFNO	0.139	8.77×10^{-5}	0.155	0.452	0.645

7.3. Runtime and speedup

Table 5 reports per-sample timings from a dedicated benchmarking pass, following the protocol of Section 4.6: GPU inference time per sample for each surrogate (CUDA, fp32), CPU rollout time per sample for each reference solver (rollout only, excluding bathymetry and source generation; float64), and their ratio as the speedup. The direct surrogates infer a full 50-frame trajectory in roughly 0.17 ms (CNN) to 3.89 ms (U-FNO), against CPU solver rollouts of 16–72 s per sample. The resulting speedups are large, from about 8.5×10^3 for U-FNO against the hydrostatic solver up to about 1.9×10^5 for the CNN against the same solver.

These ratios should be read as an order-of-magnitude, implementation-level comparison, not as a claim against optimized production tsunami codes. As stressed in Section 4.6, the reference solvers are single-threaded CPU NumPy implementations chosen to be reproducible and auditable, not tuned for speed; a vectorized, compiled, or GPU solver would shrink the denominator substantially. The point is that even against an unoptimized baseline, the surrogate moves a per-scenario forecast from tens of seconds to milliseconds or less, which is the regime change that makes large ensemble or many-scenario studies practical. Among the FNO same-solver rows the speedup also tracks the cost of the target solver: the base FNO is fastest relative to MUSCL-HR ($\approx 7.99 \times 10^4$), whose rollout is the most expensive (72.34 s), and slowest relative to Boussinesq ($\approx 1.82 \times 10^4$), whose rollout is the cheapest (16.48 s).

Table 5: Per-sample runtime and speedup. Surrogate inference is full-trajectory GPU time (CUDA, fp32); solver rollout is CPU time (float64) over 8 scenarios with 3 repeats, excluding bathymetry/source generation. Speedup is solver time divided by inference time, an implementation-level comparison with unoptimized CPU references (Section 4.6).

Model	Reference solver	Infer (ms)	Solver (s)	Speedup
FNO	Hydrostatic	0.904	33.06	$36,564\times$
FFNO	Hydrostatic	1.664	33.06	$19,873\times$
CNN	Hydrostatic	0.171	33.06	$192,921\times$
U-Net	Hydrostatic	0.237	33.06	$139,502\times$
FNO-modes8	Hydrostatic	0.861	33.06	$38,414\times$
FNO-modes20	Hydrostatic	0.954	33.06	$34,649\times$
U-FNO	Hydrostatic	3.895	33.06	$8,490\times$
WNO	Hydrostatic	2.086	33.06	$15,849\times$
FNO	MUSCL-HR	0.905	72.34	$79,933\times$
FNO	Boussinesq	0.906	16.48	$18,197\times$

7.4. Distribution-shift generalization

Table 6 reports the hydrostatic FNO scored on the test pool grouped by source family. Following Section 4.5, these are per-sample-then-grouped metrics, so the relative L_2 of a group is the mean of the per-sample values and is not directly comparable cell-for-cell with the globally aggregated number in Table 2. All six families are present in the training distribution, so this is a difficult-subgroup analysis rather than a held-out-family test.

Table 6: Source-family breakdown for the hydrostatic FNO (per-sample-then-grouped, Section 4.5). MAE is denormalized benchmark-scale; rel- L_2 is dimensionless; max-err is in normalized target units. rel- L_2 and max-err are per-sample means within the group; n is the group size.

Source family	MAE	rel- L_2	max-err	n
okada-like	0.00112	0.228	3.27	447
multi-gauss	0.00123	0.232	4.75	414
gaussian	0.00059	0.246	2.98	426
fault	0.00072	0.253	3.30	402
dipole	0.00065	0.254	3.13	408
rough	0.00090	0.344	3.48	403

The FNO degrades gracefully across most families: five of the six sit in a narrow 0.23–0.25 band, close to the in-distribution level. The clear outlier is the *rough* family (0.344), where the error is roughly 50% higher than the easiest family. This is consistent with the spectral inductive bias: rough sources inject high-wavenumber content that the truncated low-mode spectral path cannot represent as well, so the hardest family for the FNO is plausibly the one dominated by the modes it discards, though we do not isolate this with an ablation. We note this as a property to watch rather than a failure, the worst-case family is still within the same order as the in-distribution error.

We additionally evaluate three named shift suites for the hydrostatic FNO: the **multi-gauss** subset, the **trench** subset, and the high source-strength tail ($s \geq 0.82$). These give per-sample-mean relative L_2 of 0.232 and 0.267 for the first two; within the high source-strength tail the per-family error spans 0.247–0.345, with the *rough* family again the worst (0.345) and the others between 0.247 and 0.280. As in the same-solver breakdown, rough sources are the hardest case and sit above the in-distribution band, while the remaining families stay within or just above it. We stress that these suites are carved from the in-distribution test pool by metadata filtering, not held out from training; following the distinction in Section 4.5 they are therefore *difficult-subgroup* tests, not *held-out-family* tests, and are evidence of robustness within the training distribution rather than of generalization to an unseen family.

Table 7 reports the stricter held-out-family stress tests. In each row the named family is removed before preprocessing, normalization statistics are refit from the filtered training split only, and the trained FNO is evaluated both on the in-distribution remainder and on the excluded family. The result is mixed rather than solved OOD. Holding out **trench** is mild ($1.08\times$ degradation over the filtered in-distribution test), **continental** and **okada-like** are moderate ($1.47\times$ and $1.30\times$), and **rough** sources are the clear failure mode ($2.43\times$). The full-distribution model is included as a reference on the same held-out subset; it is usually better than the strict model on source holdouts, confirming that seeing the family during training matters.

Table 7: Strict held-out-family hydrostatic FNO stress tests. ID is the filtered in-distribution test remainder for the model trained without the named family; held-out is the excluded family; full model is the main FNO evaluated on the same held-out subset. Relative L_2 values are dimensionless.

Held-out family	ID rel- L_2	Held-out rel- L_2	Full model rel- L_2	Ratio
trench bathymetry	0.255	0.274	0.268	1.08
continental bathymetry	0.234	0.346	0.259	1.47
rough source	0.236	0.571	0.453	2.43
okada-like source	0.236	0.308	0.247	1.30

7.5. Cross-resolution transfer

Table 8 reports the *proxy* resolution-transfer diagnostic of Section 4.5: the same hydrostatic FNO weights are applied to the test fields resized to a target grid, with no new simulation. Because the retained Fourier-mode count is fixed while the FFT grid is rebuilt from the input size (Section 5.1), the trained operator can be evaluated at a different resolution without modification.

Table 8: Proxy resolution-transfer for the hydrostatic FNO ($N = 2500$). The same weights are evaluated on test fields resized to each grid. MAE/RMSE are denormalized benchmark-scale metrics; rel- L_2 is dimensionless; max-err is in normalized target units. The 64×64 row is the native training resolution and matches Table 2.

Grid	MAE	RMSE	rel- L_2	max-err
64×64 (native)	0.00087	0.00227	0.235	22.9
32×32 (proxy)	0.00197	0.00404	0.420	24.7

The operator runs at 32×32 without error and produces a coherent field, confirming the resolution-flexibility property. Accuracy degrades on the coarser grid (relative L_2 rising from 0.235 to 0.420), which is expected: downsampling to 32×32 removes the high-wavenumber content that the field carries at 64×64 . This proxy is therefore only a cheap resizing diagnostic, not a like-for-like simulation-resolution result. No proxy 128×128 row is reported because the current result archive contains only the 32×32 resized proxy diagnostic; the 128×128 setting is covered by the native-resolution diagnostics below. This is exactly why Section 4.5 separates the proxy from the *native* setting, where the wavefield is physically resolved at each resolution.

Table 9 reports the complementary native-resolution experiment: separate hydrostatic FNOs are trained and evaluated at 32×32 , 64×64 , and 128×128 on independently simulated fields at those grids. The split sizes are deliberately matched across resolutions, with 700 training, 150 validation, and 150 test scenarios in each native dataset, and the scenario IDs are shared across the three grids. Under this smaller-data protocol the dimensionless relative L_2 is broadly flat, 0.35–0.38, so the FNO can train and run natively at all three resolutions without a clear monotone degradation with grid size. A small data-quality caveat applies only to this native-resolution auxiliary set: during generation, the quality policy was relaxed from fail to warn for a few flagged cases so the manifest could be completed, affecting about 3 samples at 32×32 and 19 samples at 128×128 . We therefore treat the native-resolution table as a useful diagnostic, not as a cleaner dataset than the main 64×64 benchmark where all samples have `quality_status = ok`.

Table 9: Native-resolution hydrostatic FNO accuracy. Each row uses a separate model trained and evaluated at the listed grid on the matched 700/150/150 train/validation/test split. We report only dimensionless relative L_2 because absolute MAE/RMSE scales are not comparable across these native grids.

Grid	Test scenarios	rel- L_2
32×32	150	0.3792
64×64	150	0.3799
128×128	150	0.3524

These native numbers should not be compared directly to the main 64×64 FNO in Table 2 as though only the resolution changed. The native 64×64 model here uses 700 training scenarios and scores 0.3799, whereas the main hydrostatic FNO uses 10,000 training scenarios and scores 0.235; the gap is therefore a data-size/protocol difference, not evidence that the 64×64 resolution itself regressed. Likewise, the native-resolution `metrics.json` files include MAE and RMSE, but we do not rank them across grids. The stored target standardization constants are shared across the three native datasets, yet the realized denormalized target RMS on the held-out split differs substantially (0.0817, 0.0104, and 0.2014 for 32×32 , 64×64 , and 128×128), so absolute denormalized benchmark-scale or normalized-unit errors are scale-dependent. The dimensionless relative L_2 is the appropriate cross-resolution summary here.

We also tested whether the native models transfer zero-shot between native grids, using the same checkpoints as Table 9 and only changing the target test set. Table 10 shows that this is much harder than training and evaluating at the same grid. The diagonal entries reproduce the same-resolution results, while off-diagonal entries rise to 0.71–2.28. The adjacent upward case $64 \rightarrow 128$ is the best off-diagonal transfer (0.714), but it is still roughly twice the same-resolution 128×128 error. Thus the current evidence supports native training at each grid, but not a strong claim that one native-resolution checkpoint generalizes cleanly to another physically simulated grid.

Table 10: Zero-shot native-resolution transfer for hydrostatic FNO checkpoints. Rows give the training grid and columns the native evaluation grid; all checkpoints use the matched 700-train native protocol, distinct from the main 10,000-train 64×64 FNO. Entries are relative L_2 on the 150-scenario test split.

Train grid	Train N	Test 32×32	Test 64×64	Test 128×128
32×32	700	0.379	0.956	0.911
64×64	700	1.212	0.380	0.714
128×128	700	2.282	1.937	0.352

7.6. Synthetic-to-real bathymetry transfer

The main real-bathymetry diagnostic uses a 10-crop GEBCO-derived morphology suite covering fully wet offshore regions including Sumatra–Java, Alaska–Aleutian, Cascadia, the Hellenic Arc/Aegean, Florida–Bahamas, the Caribbean, the Philippines east trench, Kuril–Japan, Tohoku offshore, and Nankai. The real-bathymetry diagnostic crops are derived from the GEBCO_2026

Grid, a global ocean-and-land terrain model provided at 15 arc-second resolution [7]; after cropping, they are resampled to the model’s 64×64 grid and linearly rescaled to the fully wet training range $[-10, -0.75]$. The raw files contain bathymetry only, so each crop is paired with a synthetic initial-displacement source, advanced with the hydrostatic solver, and preprocessed with the main hydrostatic training normalization. This is therefore a synthetic-source/real-bathymetry morphology-transfer suite, not real-event validation.

Table 11: Synthetic-source transfer on the 10-crop fully wet real-bathymetry morphology suite. Direct models use static inputs; windowed models are first-target-frame-seeded diagnostics scored on the remaining 49 predicted target frames, so they are not like-for-like with direct 50-frame prediction. Relative L_2 is shown in normalized and denormalized benchmark-scale forms.

Model	Samples	rel- L_2	denorm. rel- L_2
Direct FNO	10	0.275	0.271
Direct FFNO	10	0.183	0.179
Window-FNO-5	10	0.053	0.052
Window-FFNO-5	10	0.039	0.038

FFNO improves the direct FNO on this suite from $\text{rel-}L_2 = 0.275$ to 0.183. The windowed scores are lower (Window-FNO-5 0.053, Window-FFNO-5 0.039), but they use the first-target-frame-seeded diagnostic of Section 7.8, are scored on the remaining 49 predicted target frames, and are not like-for-like with direct 50-frame prediction. We therefore claim only transfer to fully wet real bathymetric morphologies under synthetic forcing, not real-event or site-specific hazard skill. Figure 4 shows one illustrative Tohoku offshore crop; the coastline variants are in Appendix A.

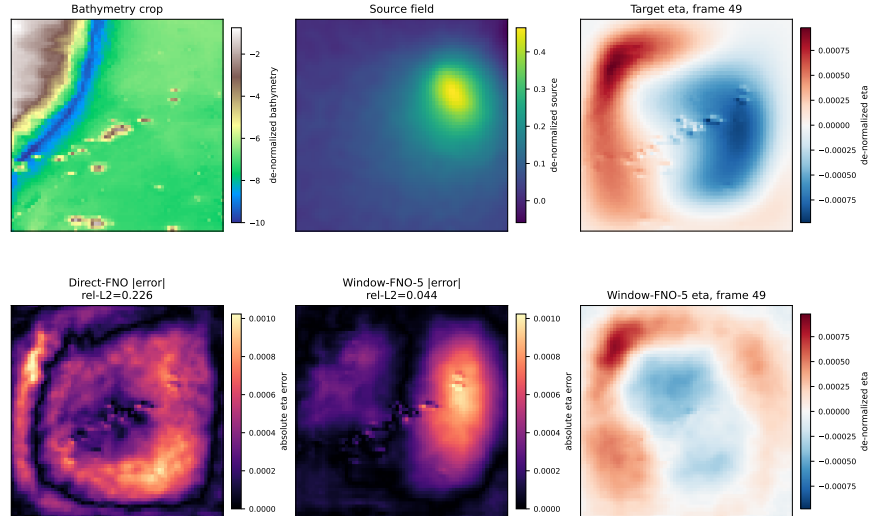


Figure 4: Synthetic-forcing morphology transfer on one GEBCO-derived Tohoku crop rescaled to $[-10, -0.75]$. Top: bathymetry, synthetic source, and hydrostatic reference η at frame 49. Bottom: direct-FNO error, Window-FNO-5 error, and Window-FNO-5 η on the reference scale. Panel rel- L_2 labels are single-crop trajectory scores for this illustrative case, not the 10-crop aggregate in Table 11; this is not real-event validation.

7.7. Solver-gap and cross-solver diagnostics

Because the training solver is not itself ground truth, Section 4.7 separates two quantities: the raw denormalized benchmark-scale gap between two numerical references on shared scenarios, and the surrogate error of a model trained on one reference and scored against another. Table 12 reports the raw solver-vs-solver gaps on all 2,500 shared test scenarios, in both directions for each pair. These are same-scenario, same-saved-index diagnostics on the released labels, not strict common-physical-time solver-fidelity comparisons. The two shallow-water references are close: hydrostatic versus MUSCL-HR has a mean relative L_2 of 0.164 (RMSE 0.00173). Both shallow-water references are far from the weakly dispersive Boussinesq comparison reference: hydrostatic versus Boussinesq has a mean relative L_2 of 0.861 and MUSCL-HR versus Boussinesq 0.897, an order of magnitude larger in RMSE (≈ 0.016 – 0.017). The reverse Boussinesq-vs-shallow-water directions give larger relative L_2 (1.60 and 1.55). We interpret this only as directional normalization: each row divides the same symmetric difference by the norm of the row’s reference field, so the two directions answer different “relative to which reference?” questions. The symmetric RMSE and MAE are therefore the safer scale comparisons. This gap structure is consistent with the different numerical references, but because Boussinesq has a different saved-frame physical-time scale, we do not read the cross-family rows as strict common-physical-time fidelity errors.

Table 12: Raw solver-vs-solver gaps on shared test scenarios ($N = 2500$, field η trajectory), compared as same-scenario, same-saved-index released labels. RMSE and MAE are denormalized benchmark-scale metrics; relative L_2 is a per-sample mean. The relative L_2 is asymmetric because it normalizes by the second solver’s field norm; RMSE and MAE are symmetric.

Solver pair (A vs B)	MAE	RMSE	rel- L_2
Hydrostatic MUSCL-HR	0.00084	0.00173	0.164
MUSCL-HR Hydrostatic	0.00084	0.00173	0.177
Hydrostatic Boussinesq	0.00651	0.01632	0.861
Boussinesq Hydrostatic	0.00651	0.01632	1.603
MUSCL-HR Boussinesq	0.00692	0.01690	0.897
Boussinesq MUSCL-HR	0.00692	0.01690	1.548

Emulator-superiority ratios. Table 13 reports the emulator-superiority ratio of Section 4.7,

$$\rho(A \rightarrow B) = \frac{\text{RMSE}(f_{\theta}^A, \text{ref } B)}{\text{RMSE}(\text{ref } A, \text{ref } B)}.$$

The numerator is the denormalized benchmark-scale RMSE of a surrogate trained on reference A and scored against the targets of a *different* reference B ; the denominator is the raw A -vs- B solver gap. Because all three references are generated on the same scenarios with identical input channels (Section 7.1), a surrogate can be evaluated against another reference’s targets without distribution shift in its inputs. A value $\rho < 1$ would mean the surrogate is closer to B than A ’s own solver is.

Across all four cross-reference pairs the ratio exceeds one. Within the shallow-water family the hydrostatic-trained FNO scored against MUSCL-HR gives $\rho = 1.98$ and the MUSCL-HR-trained FNO against hydrostatic gives $\rho = 1.76$: the two shallow-water solvers already agree closely (Table 12, gap 0.00173), so the surrogate’s own emulation error is larger than the small

reference gap it would have to beat. For the cross-family pairs the Boussinesq-trained FNO scored against the hydrostatic and MUSCL-HR targets gives $\rho = 1.33$ in both cases: although the raw cross-family solver gap is an order of magnitude larger (≈ 0.016), the Boussinesq surrogate’s error *on the shallow-water targets* (≈ 0.022) is larger still, because those targets are governed by physics it was not trained on. The consistent reading is that the emulator is not “superior” to the numerical reference in any of these pairs: a surrogate reproduces the solver it was trained on far better than it reproduces a different solver, and the cross-solver reference gap is not something the present surrogate closes. We regard this as a negative diagnostic outcome rather than a shortfall of the method.

Table 13: Emulator-superiority ratios from Section 4.7. RMSE_f is the cross-reference surrogate error and RMSE_{AB} is the raw solver-gap denominator from Table 12. Values $\rho < 1$ would indicate surrogate superiority; all rows give $\rho > 1$.

Surrogate (trained on A)	vs ref B	RMSE_f	RMSE_{AB}	ρ
FNO (Hydrostatic)	MUSCL-HR	0.00343	0.00173	1.98
FNO (MUSCL-HR)	Hydrostatic	0.00305	0.00173	1.76
FNO (Boussinesq)	Hydrostatic	0.02174	0.01632	1.33
FNO (Boussinesq)	MUSCL-HR	0.02241	0.01690	1.33

Normalization check. The cross-reference diagnostic checks both the target denormalization signature and the input-channel normalization statistics between the prediction dataset and the reference dataset. For all four pairs the input-channel signature check passes: the bathymetry and source channels share identical normalization statistics across the prediction and reference datasets (the `initial_depth` channel is in stored scale on both sides), so the surrogate sees inputs on the same scale it was trained on. The target denormalization is applied separately to prediction and reference using each dataset’s own stored statistics, so the denormalized benchmark-scale numerator and denominator are each well defined in their native scale. The ratios are therefore reported as direct denormalized benchmark-scale comparisons, and the qualitative conclusion ($\rho > 1$ throughout) is unaffected.

7.8. Windowed autoregressive forecaster

To target the per-frame drift of the single-pass models (Table 3), we train windowed FNO and FFNO variants on the hydrostatic reference that forecast the trajectory autoregressively in blocks of $K = 5$ frames: from an input stack $[\text{bathymetry}, \text{source}, \eta_t, \eta_{t-1}]$ each model predicts the next five frames, then re-conditions on its own last two predicted frames and advances, covering the 50-frame horizon in ten chunks. This trades one long-horizon prediction for several short-horizon ones at the cost of an autoregressive feedback loop.

Table 14 compares the windowed models against their direct single-pass counterparts on the hydrostatic test split. The windowed rows are first-target-frame-seeded diagnostics scored on the remaining 49 predicted target frames, not like-for-like direct 50-frame predictions. Window-FNO-5 reaches a full-trajectory relative L_2 of 0.077, against 0.235 for the single-pass FNO. Window-FFNO-5 is lower again at 0.052 (denormalized benchmark-scale $\text{rel-}L_2 = 0.0513$), compared with 0.168 for

the direct FFNO. The same drift pattern remains: the windowed FFNO starts at 0.030 and rises to 0.169 by the final predicted frame, so it reduces but does not eliminate temporal accumulation.

Table 14: Windowed versus direct hydrostatic models on the test split ($N = 2500$). Params are in millions; full-traj is global relative L_2 ; per-frame columns are relative L_2 at the first scored prediction, mid horizon, and final frame. Direct models predict all 50 target frames from static inputs; windowed models are first-target-frame-seeded diagnostics scored on the remaining 49 predicted target frames, so the tasks are not like-for-like. Inference time is per sample on GPU.

Model	Params (M)	full-traj	first scored	mid	final	infer (ms)
Direct FNO	5.443	0.235	0.214	0.325	0.309	0.90
Direct FFNO	0.411	0.168	0.135	0.264	0.273	1.66
Window-FNO-5	5.439	0.077	0.051	0.130	0.172	11.1
Window-FFNO-5	0.407	0.052	0.030	0.105	0.169	21.6

Two caveats are essential for a fair reading. First, the comparison is not strictly like-for-like: the windowed rollout is a *first-target-frame-seeded diagnostic* and its metrics are computed over the remaining 49 predicted target frames, whereas the direct models predict all 50 target frames from the static inputs alone with no seed. Part of the windowed models’ advantage at early frames therefore reflects the easier, partially observed task; the late-frame improvement is more informative but still not a clean direct-task comparison. Second, the windowed models are slower because they run ten sequential forward passes: Window-FNO-5 costs 11.1 ms and Window-FFNO-5 costs 21.6 ms per sample. Window-FFNO-5 is much smaller than Window-FNO-5 (0.407M vs 5.439M parameters), but it is slower in this implementation, so parameter count should not be read as measured throughput. Even so, they remain orders of magnitude faster than the CPU hydrostatic reference. We also do not compute the emulator-superiority ratio for the windowed models, because their seeded numerator would be a different task from the single-pass surrogate used in Section 7.7.

On the 10-crop real-bathymetry morphology suite of Section 7.6, the same seeded pattern holds: Window-FNO-5 reaches $\text{rel-}L_2 = 0.053$ and Window-FFNO-5 reaches 0.039, compared with 0.275 and 0.183 for their direct counterparts. These are again first-target-frame-seeded diagnostics scored on the remaining 49 predicted target frames, not like-for-like direct 50-frame predictions. In the appendix coastline ablation, the fully wet coastline remains easy for the windowed models, while the wet-dry version fails, so we treat the wet-dry stress result as a limitation rather than evidence for the main transfer claim. We additionally score Window-FNO-5 on the same distribution-shift and native-resolution suites used for the single-pass FNO, with the same first-target-frame-seeded caveat applied throughout. The native-resolution result differs: Window-FNO-5 is trained only at 64×64 , so evaluating the same checkpoint at other native grids is a zero-shot resolution transfer. At its training resolution it scores 0.076 (in line with the main test split), but it fails when transferred zero-shot to 32×32 (1.101) and 128×128 (0.828), mirroring the single-pass zero-shot transfer collapse in Table 10. We therefore report the windowed cross-resolution result only as a same-resolution number plus an explicit zero-shot-transfer failure, not as a native-resolution training claim. Full per-suite numbers are tabulated in the appendix (Table 27).

7.9. Uncertainty

We quantify predictive uncertainty with a deep ensemble of $M = 7$ hydrostatic FNOs that differ only in random seed (seeds 11, 22, 33, 44, 55, 66, 77), following the protocol of Section 4.8. Table 15 reports two complementary raw diagnostics: the empirical coverage of uncalibrated central Gaussian prediction intervals at four nominal levels, and the Pearson correlation between the per-element predictive standard deviation and the absolute error, on the in-distribution test split and on the three distribution-shift suites.

Table 15: Ensemble uncertainty diagnostics for the hydrostatic FNO ($M = 7$). Coverage is raw empirical central-interval coverage before scalar calibration (ideal 0.50/0.80/0.90/0.95); err-unc. is the Pearson correlation between predictive standard deviation and absolute error. The first row is the full $N = 2500$ test split; others are distribution-shift suites.

Suite	cov ₅₀	cov ₈₀	cov ₉₀	cov ₉₅	err-unc.
In-distribution	0.463	0.700	0.782	0.832	0.646
multi-gauss subset	0.398	0.630	0.721	0.780	0.658
trench subset	0.488	0.725	0.804	0.851	0.648
high source-strength tail	0.376	0.589	0.673	0.728	0.700

Two findings stand out. First, the raw ensemble is *overconfident*: empirical coverage remains below the nominal level at every interval, with the nominal 95% interval capturing 83% of in-distribution targets and 73–85% under shift. Thus the uncalibrated Gaussian intervals built from ensemble variance are too narrow and should not be read as calibrated probabilities. Second, and more positively, the error–uncertainty correlation is stable and moderately strong across all settings (0.65–0.70), and is highest on the high source-strength tail (0.700). So even though the raw scale of the uncertainty is miscalibrated, its *spatial structure* tracks where the model is actually wrong, which is the property that matters for flagging untrustworthy regions of a prediction.

The behavior under shift is mixed. Raw coverage does not improve uniformly under distribution shift, and on the high source-strength tail the normalized Gaussian negative log-likelihood is worse than in distribution (−0.22 versus −1.91), meaning the variance still fails to grow enough to match the larger errors in that tail. The **trench** bathymetry suite is the one raw case where coverage edges above the in-distribution level, but not enough to reach the nominal targets. Appendix A.3 reports the full reliability/calibration diagnostic: a validation-only scalar variance inflation ($\gamma = 3.359364$) improves marginal coverage on the in-distribution test split and some OOD suites, while the hardest high source-strength tail remains under-covered. We therefore treat these as reliability diagnostics rather than calibrated risk estimates: the ensemble produces a useful relative ranking of where error concentrates, and the post-hoc correction improves interval scale, but per-element marginal coverage over correlated pixels and frames is not operational risk calibration.

8. Discussion

8.1. Main findings

Across the benchmark configurations tested here, FFNO is the strongest single-run direct hydrostatic configuration tested, while the base FNO remains competitive across all three numerical references (Table 2). The implementation-level speedups over the unoptimized CPU reference solvers are several orders of magnitude, and for the FNO same-solver rows they track the cost of the target solver (Table 5). The single-pass models’ main weakness is per-frame drift over the forecast horizon (Table 3), which the seeded autoregressive windowed forecasters directly target: Window-FFNO-5 reaches a full-trajectory error of 0.052 as a first-target-frame-seeded diagnostic scored on the remaining 49 predicted target frames, at the cost of sequential inference and a task that is not like-for-like with direct 50-frame prediction (Section 7.8). Robustness is mixed but interpretable: metadata-filtered source-family diagnostics are mostly close to the in-distribution band, strict held-out-family stress tests range from mild trench degradation to a rough-source failure, and zero-shot transfer between natively simulated resolutions collapses for both the single-pass and windowed models. The deep ensemble gives a usable but overconfident uncertainty (Section 7.9): the error–uncertainty correlation is stable around 0.65–0.70 in- and out-of-distribution, while interval coverage still falls short of nominal, so the uncertainty is best read as a relative ranking signal rather than a calibrated probability.

8.2. Effect of reference choice

The two shallow-water references behave almost interchangeably from the surrogate’s point of view. Their raw solver-vs-solver gap is small (relative L_2 0.164, RMSE 0.00173; Table 12), and the FNO trained on either one reaches a similar same-solver accuracy (0.235 hydrostatic, 0.263 MUSCL-HR). The Boussinesq reference is the genuinely different target: its weakly dispersive correction puts it an order of magnitude away from both shallow-water solvers in denormalized benchmark-scale RMSE (≈ 0.016 – 0.017), and that cross-family gap dwarfs the within-family one. These are same-scenario, same-saved-index released-label diagnostics rather than strict common-physical-time solver-fidelity measurements. This separation is exactly what motivates a multi-reference benchmark: a surrogate that looks solver-agnostic within the shallow-water family is in fact being tested against a near-identical target, and only the dispersive reference exposes how much of the learned behavior is tied to the training physics. The cross-solver diagnostic does *not* support any emulator-superiority-like behavior: the emulator-superiority ratio exceeds one in all four cross-reference pairs (Table 13), from 1.76–1.98 within the shallow-water family to 1.33 across families. A surrogate reproduces the solver it was trained on far better than it reproduces a different solver, and it does not close the reference gap between solvers. We read this as a negative diagnostic outcome: the surrogate is an emulator of its training solver, not a more accurate physics model than that solver.

8.3. Model behavior

The accuracy ranking of the direct models is consistent with the inductive-bias argument of Section 5. The wavefields here are governed by long-range propagation: a disturbance at the source influences distant cells within the forecast horizon, so a model that can move information globally in one layer has an advantage over one that must build up an effective receptive field through depth. The FFNO and FNO spectral paths provide that global coupling, the encoder–decoder U-Net reaches intermediate range through downsampling, and the plain CNN is purely local; the measured ordering among the tested single-run configurations follows that reasoning. We present this as a plausible interpretation rather than a demonstrated mechanism. FFNO’s direct hydrostatic improvement is not due to a larger parameter budget: it has 0.411M parameters versus 5.443M for the base FNO. The mode ablations also show that simply increasing retained Fourier modes from 14 to 20 is not enough to improve the base FNO. Where the global modeling helps least is the high-wavenumber regime: the rough source family is the hardest strict source holdout, consistent with the truncated spectral path struggling with content those sources inject. The shared failure across architectures is temporal drift in the single-pass setting and a collapse under zero-shot native-resolution transfer, neither of which is specific to the spectral model; the windowed rollout addresses the first but not the second.

8.4. Boussinesq comparison reference

The Boussinesq reference is included because it provides a target that the shallow-water surrogates do not reproduce as closely. In dimensionless relative L_2 the Boussinesq FNO looks strongest (0.210), and its per-frame error is essentially flat over the horizon (0.207 $\rightarrow$ 0.208). This ranking is unit-dependent: the Boussinesq target normalization scale is about $2.3\times$ larger, so in denormalized benchmark-scale surface-elevation units the Boussinesq surrogate is the *least* accurate by RMSE (0.00468 vs 0.00227 hydrostatic), and its flatness reflects the smoothness of the weakly dispersive elevation field rather than a better model. The comparison therefore supplies a controlled weakly dispersive contrast, still a synthetic NumPy reference with health checks rather than a validated truth model, and shows that the cross-family released-label gap is not closed by the surrogate. We keep hydrostatic reconstruction as the primary reference and treat MUSCL-HR and Boussinesq as comparison baselines, so the Boussinesq result is a same-saved-index reference-contrast diagnostic, not a claim that the dispersive solver is closer to nature.

8.5. Practical implications

The benchmark is a controlled study of surrogate behavior, not an operational tsunami forecasting tool or site-specific hazard assessment. Its scope is narrower: it tests how neural surrogates of solver-generated shallow-water wavefields behave under solver mismatch, distribution shift, resolution change, synthetic-source transfer onto fully wet real-bathymetry morphologies, and uncertainty diagnostics. Fast surrogates may eventually support scenario exploration, educational tools, rapid sensitivity analysis, or decision-support workflows, but only when combined with validated physical models and real-world data. We therefore frame downstream relevance as future research rather than a capability the present benchmark already provides; the 10-crop real-bathymetry morphology suite is a modest transfer diagnostic, not evidence of operational readiness.

9. Limitations

The first limitation is that this is a controlled synthetic benchmark. Procedural bathymetry and source fields provide many wave-propagation cases, but they do not replace surveyed bathymetry, realistic earthquake rupture models, tides, bottom friction, coastal inundation geometry, or validation against observed tsunami records. The 10-crop real-bathymetry suite is fully wet and uses synthetic sources, so it tests morphology transfer outside the procedural terrain distribution, not readiness for operational tsunami warning or site-specific hazard assessment.

The labels are numerical reference solutions, not physical ground truth. The Hydrostatic and MUSCL-HR solvers are finite-volume shallow-water solvers with adaptive CFL time stepping, open boundaries, sponge damping, dry-cell protection, and quality checks for non-finite or unstable rollouts. These choices make the labels reproducible and safer for training, but they do not remove solver discretization error, boundary-treatment error, or resolution dependence, and they are not substitutes for formal convergence proof, external-solver validation, observed-event validation, or production-code validation. Therefore, an error number in this paper is an error against a stated solver and configuration, not against the real ocean. This is why we keep the raw solver-vs-solver gap separate from the surrogate-vs-surrogate error.

The Boussinesq comparison reference also has a limited interpretation. The implemented solver is an elevation-only weakly dispersive model with its own acceleration solve, filtering, sponge damping, finite-depth safeguards, and convergence checks rather than a full nonlinear production Boussinesq tsunami model. It is useful for comparison with Hydrostatic and MUSCL-HR on shared scenarios, but it should not be treated as real physical ground truth or as an automatic upgrade over the shallow-water labels.

The released trajectories are fixed saved-step-index labels, not outputs interpolated onto a common physical-time grid. The raw archives store timestamps and time-step histories, and the audit in Appendix A.2 shows that the shallow-water references are nearly aligned at saved frame 50 while Boussinesq frame 50 is at about half the shallow-water physical time. Same-solver surrogate metrics are therefore interpreted as emulation of the stored rollout sequence. Cross-solver gaps involving Boussinesq are useful same-scenario, same-saved-index label-space diagnostics, but they should not be read as strict common-physical-time solver-fidelity measurements.

The runtime and hardware comparisons are implementation-level comparisons. The solver timing scripts re-run the NumPy reference solvers on CPU from cached bathymetry/source scenarios, while the neural model can be timed on CPU or GPU with recorded batch size, precision, and TF32 settings. Dataset generation used an 8-core AMD EPYC 9B45 CPU server, while training and evaluation used an NVIDIA GeForce RTX 4050 Laptop GPU with an Intel Core i5-13420H CPU. This is the correct comparison for this repository and is representative of resource-constrained research environments, but it is not a universal speed claim against optimized production tsunami codes, GPU finite-volume solvers, adaptive-mesh solvers, or operational forecast systems. The speedup values should therefore be read as acceleration relative to our reproducible reference implementations, not as upper bounds on surrogate performance.

The OOD and resolution-transfer results depend on how the processed datasets are built. The OOD suite builder filters existing evaluation archives using metadata such as `source_type`, `bathymetry_type`, and `source_strength`. This is useful for difficult-subgroup analysis, but the

stricter held-out-family results show that removing a family before preprocessing can still produce large degradation, especially for rough sources. Similarly, proxy resolution transfer resizes processed inputs and targets, so it is a cheaper diagnostic and not the same as a newly simulated grid. Native-resolution evaluation is stronger, but only if the target normalization statistics are compatible with the training reference; otherwise a normalization mismatch can look like a resolution-generalization effect.

Finally, the paper is limited to forward emulation. The repository contains an inverse-dataset scaffold, including dense-map and sparse-gauge observation formats, but it does not yet train or validate an inverse model for source reconstruction or real-time data assimilation. Those problems need different inputs, metrics, uncertainty checks, and validation protocols, so we leave them for a separate follow-up study instead of mixing them into the forward-surrogate benchmark.

10. Conclusion

We presented a controlled forward benchmark for neural emulation of solver-generated tsunami-like wavefields. The benchmark pairs procedural bathymetry with diverse initial-displacement sources on shared scenarios, and labels each scenario with up to three numerical references – hydrostatic reconstruction as the primary shallow-water reference, MUSCL-HR as a second shallow-water reference, and an elevation-only weakly dispersive Boussinesq solver – so that the same input can be compared across reference choices. On this benchmark we studied FNO and FFNO surrogates against convolutional and operator baselines under a matched protocol, covering accuracy, runtime, distribution shift, cross-resolution transfer, synthetic-source transfer onto fully wet real-bathymetry morphologies, physics diagnostics, and predictive uncertainty.

The core findings are consistent across the diagnostics. Among the single-run direct hydrostatic configurations tested, FFNO is strongest, while the base FNO remains competitive across all three references; the Boussinesq ranking is unit-dependent and reverses in denormalized benchmark-scale RMSE. Single-pass shallow-water models drift upward over the forecast horizon, and seeded windowed variants reduce that error substantially, with Window-FFNO-5 reaching 0.052 as a first-target-frame-seeded diagnostic scored on the remaining 49 predicted target frames, not as a like-for-like direct 50-frame prediction. Inference is several orders of magnitude faster than the CPU NumPy reference implementations used here, which makes large-ensemble and many-scenario studies practical. The strict held-out-family tests are mixed, ranging from mild trench degradation to a rough-source failure, and the 10-crop real-bathymetry suite supports only a modest fully wet morphology-transfer claim under synthetic forcing. The cross-solver diagnostic returns a negative result: the emulator-superiority ratio exceeds one in every pair, so a surrogate reproduces the solver it was trained on far better than it reproduces a different solver, and it does not close the same-scenario, same-saved-index released-label gap between solvers. The deep ensemble gives an uncertainty signal that tracks error usefully (correlation ≈ 0.65 – 0.70) but is systematically overconfident, with nominal intervals under-covering both in-distribution and under shift.

We emphasize that multi-reference evaluation is what makes these readings trustworthy rather than dependent on a single label choice. Keeping the raw solver-vs-solver gap separate from the surrogate-vs-surrogate error prevents a surrogate’s emulation skill from being confused with disagreement between numerical models, and reporting every error against a stated solver and con-

figuration keeps the claims conservative: these are errors against numerical references, not against the real ocean.

Future work is broader. Real-event transfer needs surveyed bathymetry, realistic rupture models, shoreline and wet-dry physics, and validation against observed records. Runtime should be compared with optimized GPU or compiled finite-volume solvers. Full nonlinear Boussinesq references would sharpen cross-reference tests, and uncertainty estimates need recalibration and larger ensembles before any reliability claim. Source reconstruction from sparse observations needs different inputs, metrics, and validation, so we leave it to a separate follow-up study rather than mixing it into the forward-surrogate benchmark.

Acknowledgments

The authors thank Doan Minh Khoi for assistance with payment of compute resources used in this work. Dataset generation was carried out on a server provisioned through the Google Cloud free tier. The authors also acknowledge the University of Information Technology, Vietnam National University Ho Chi Minh City, for providing the academic environment in which this work was carried out.

Declarations

Funding. This work was self-funded by the authors. No specific grant from any funding agency in the public, commercial, or not-for-profit sectors was received.

Competing interests. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Author contributions (CRediT). **Nguyen Tho Binh An:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing. **Le Minh Nhut Tan:** Conceptualization, Methodology, Software, Validation, Investigation. **Mai Tien Dung:** Supervision, Conceptualization, Writing – review & editing.

Use of generative AI. During the preparation of this work the authors used generative AI tools to assist with language editing and code. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Data and code availability

The source code, configuration files, data-generation and evaluation scripts, plotting scripts, and the result files (JSON/CSV) underlying every table and figure in this paper are publicly available at <https://github.com/1zuki/tsunami-surrogate>.

The processed benchmark datasets (model-ready train/validation/test arrays for the hydrostatic, MUSCL-HR, and Boussinesq references), together with the scenario manifests, train/validation/test scenario identifiers, normalization statistics, and checksums, are archived with a persistent identifier at <https://doi.org/10.5281/zenodo.20974604>. The full raw rollout archive (≈ 300 GB) can be regenerated from the released scripts and configuration files; a mirror is additionally available at https://drive.google.com/drive/folders/16hC9hTBBTw_xygkEtc45bYrM2mjrJ5TU?usp=sharing.

The real-bathymetry diagnostic crops were derived from the GEBCO_2026 Grid [7]; we acknowledge the GEBCO Bathymetric Compilation Group 2026 and the NERC EDS British Oceanographic Data Centre NOC for providing the dataset.

Trained model checkpoints are not redistributed in full, but the released repository includes the resolved configurations, random seeds, the checkpoint-selection rule, the final per-model metrics, and the training histories needed to reproduce the reported results.

A. Additional implementation and health-check details

A.1. Configuration details

The experiments share a single base training configuration, with per-model overrides layered on top. Table 16 lists the settings that are referenced only indirectly in the main text. The main hydrostatic FNO widens the base architecture (width 48, depth 6, 14 spectral modes per axis against the base 32/4/12) and replaces the plain MSE objective with a horizon-weighted MSE that up-weights late frames, together with cosine learning-rate decay, gradient clipping, and a longer early-stopping patience. The convolutional baselines and the cross-resolution and windowed variants reuse this base and change only the fields noted in their respective sections.

Table 16: Key configuration settings for the main hydrostatic FNO, as resolved from the base training config and the model override. Output channels equal the number of predicted frames times the per-frame channel count.

Group	Setting	Value
Architecture	spectral modes (m_1, m_2)	14, 14
	width	48
	depth	6
	spatial padding	6
	coordinate grid channels	enabled
Optimization	optimizer	AdamW
	learning rate	5×10^{-4}
	min. learning rate	5×10^{-5}
	schedule	cosine
	weight decay	1×10^{-6}
	gradient clip	1.0
	loss	horizon-weighted MSE
	weight range / power	0.8–2.0 / 1.7
	max epochs	256
	early-stopping patience	20
	checkpoint metric	val. rel- L_2
Data	batch size	64
	train / val / test pool	10,000 / 1,000 / 2,500
Ensemble	member seeds	{11, 22, 33, 44, 55, 66, 77}

A.2. Additional solver diagnostics

Beyond the dataset-level quality policy of Section 3, both shallow-water references are checked against two standard finite-volume sanity properties in the test suite. The first is the lake-at-rest (well-balanced) property: starting from a flat free surface $\eta_0 = 0$ over non-flat fully submerged bathymetry with zero initial momentum, a well-balanced scheme must keep the surface flat and the momentum zero rather than spuriously generating flow at the bathymetry gradients. Both the hydrostatic and MUSCL-HR solvers hold this to a tight tolerance over a short rollout: the maximum free-surface drift stays below 10^{-8} and the maximum momentum component below 10^{-7} in normalized units. The second is a dynamic-pulse stability check: a small Gaussian surface perturbation is advanced over flat, sloped, and multi-Gaussian bathymetry under periodic and open boundaries, and the rollout must stay finite, keep depth nonnegative (to within 10^{-8}), respect the velocity safeguard, and still produce a non-trivial propagating signal. The MUSCL-HR solver passes these across the tested grids (32^2 and 64^2) and boundary conditions. These are sanity checks, not convergence-order measurements, external-solver validations, observed-event validations, or production-code validations; they are guardrails confirming that the references preserve the trivial steady state and remain stable on dynamic inputs, which is what justifies treating their outputs as clean supervised-learning labels.

Saved-frame timestamp audit. The raw archives store `timestamps` and `dt_history`, while the processed arrays retain `sample_dir` pointers back to those raw files. We audited all processed metadata rows without rerunning the dataset. Table 17 reports the physical timestamp attached to saved frame index 50, which corresponds to solver update 250 under the default “include initial state, save every 5 steps” policy. Hydrostatic and MUSCL-HR are nearly aligned at this saved index. Boussinesq is not: its frame 50 timestamp is about half the shallow-water timestamp. The released benchmark therefore uses fixed saved-step/index alignment, not interpolation to common physical times.

Table 17: Saved-frame timestamp audit over all processed train/validation/test rows (13,500 per solver). Pairwise rows summarize same-split, same-scenario absolute timestamp gaps at saved frame index 50.

Quantity	Item	Mean	Range / Max
Frame-50 timestamp	Hydrostatic	0.177402	[0.176231, 0.177496]
Frame-50 timestamp	MUSCL-HR	0.177390	[0.175959, 0.177517]
Frame-50 timestamp	Boussinesq	0.088738	[0.088738, 0.088738]
$ t_{50,A} - t_{50,B} $	Hydrostatic–MUSCL-HR	1.28×10^{-5}	2.82×10^{-4}
$ t_{50,A} - t_{50,B} $	Hydrostatic–Boussinesq	8.87×10^{-2}	8.88×10^{-2}
$ t_{50,A} - t_{50,B} $	MUSCL-HR–Boussinesq	8.87×10^{-2}	8.88×10^{-2}

Standard SWE sanity cases. We additionally ran three shallow-water sanity cases at 128×128 , CFL 0.40, reflective no-sponge boundaries, and matched final time 0.05: a radial Gaussian pulse over flat bathymetry, a Gaussian pulse over smooth variable bathymetry, and a wet-bed dam-break. Table 18 summarizes the checks. Both shallow-water solvers remain finite, keep depth positive, conserve mass to roundoff, preserve the expected symmetries, and pass all six solver-case checks. These are standard SWE sanity checks, not formal validation against observed events or external production solvers.

Table 18: Standard SWE validation cases for the Hydrostatic and MUSCL-HR solvers on the full diagnostic run (grid 128, CFL 0.40, final time 0.05). Mass drift and extra metrics are worst values over the two solvers.

Case	Max mass rel. drift	Extra diagnostic	Result
Flat radial Gaussian	0	symmetry error $\leq 1.11 \times 10^{-15}$; min. depth 0.9877	2/2 pass
Smooth-bathymetry Gaussian	Gaus- 0	min. depth 1.038	2/2 pass
Wet-bed dam-break	1.11×10^{-16}	y -invariance error 0; min. depth 0.95	2/2 pass

Minimal grid/time sensitivity. For a lightweight grid/time diagnostic, we generated 12 smooth shared scenarios at 128×128 , restricted them to 64×64 and 32×32 by cell averaging, and advanced Hydrostatic and MUSCL-HR to final time 0.006 with the last adaptive step clipped. The full diagnostic uses the same three grids, CFL values 0.30 and 0.45, and two shallow-water solvers, giving 144 rollouts. Table 19 reports final- η comparisons after restricting the finer-grid result to the coarser grid. The errors generally shrink from the 32–64 comparison to the 64–128

comparison, and the CFL 0.30 versus 0.45 final fields remain close (mean rel- $L_2 \leq 9.79 \times 10^{-3}$ for Hydrostatic and $\leq 5.08 \times 10^{-3}$ for MUSCL-HR across grids). This documents stability and sensitivity on shared scenarios; it is not a formal convergence proof, external-solver validation, or production-code validation.

Table 19: Minimal shared-scenario grid sensitivity. Values are mean final- η relative L_2 over 12 scenarios at matched final physical time 0.006.

Solver	CFL	32 vs restricted 64	64 vs restricted 128
Hydrostatic	0.30	5.04×10^{-2}	2.74×10^{-2}
Hydrostatic	0.45	5.11×10^{-2}	2.41×10^{-2}
MUSCL-HR	0.30	3.95×10^{-2}	1.45×10^{-2}
MUSCL-HR	0.45	4.22×10^{-2}	1.57×10^{-2}

Table 20: CFL sensitivity at matched rollout end time. Values compare CFL 0.30 against CFL 0.45 over the same 12 scenarios.

Solver	Grid	Mean final rel- L_2	Max final rel- L_2	Mean RMSE
Hydrostatic	32	9.79×10^{-3}	1.31×10^{-2}	3.63×10^{-5}
Hydrostatic	64	8.56×10^{-3}	1.19×10^{-2}	3.29×10^{-5}
Hydrostatic	128	4.83×10^{-3}	6.72×10^{-3}	1.90×10^{-5}
MUSCL-HR	32	5.08×10^{-3}	9.60×10^{-3}	1.85×10^{-5}
MUSCL-HR	64	2.02×10^{-3}	4.22×10^{-3}	7.54×10^{-6}
MUSCL-HR	128	4.83×10^{-4}	1.02×10^{-3}	1.82×10^{-6}

CFL 0.30 and 0.45 remain bounded in this diagnostic; the table reports sensitivity only between those two operating settings.

Boussinesq dispersion check. As an appendix-only sensitivity check for the weakly dispersive reference, we propagated small-amplitude sinusoidal waves on a 256×32 constant-depth grid with periodic boundaries, no sponge, and no filter. The full diagnostic uses modes 1, 2, 4, 8, 12, 16, 24, 32, final time 1.5, and fixed step size $\Delta t = 1.25 \times 10^{-4}$. The initial condition uses the expected discrete frequency for a right-going Fourier mode, and the measured frequency is recovered from the unwrapped phase of the corresponding Fourier coefficient. The expected trend is the implemented discrete relation

$$\omega^2 = \frac{gHk_d^2}{1 + \alpha_B H^2 k_d^2}, \quad k_d = \frac{2 \sin(k\Delta x/2)}{\Delta x}.$$

Figure 5 shows the measured phase speeds for $\alpha_B = 0$ and $\alpha_B = 1/3$. With $\alpha_B = 0$, the phase speed follows the non-dispersive shallow-water value modified only by the discrete wavenumber. With $\alpha_B = 1/3$, the higher modes slow sharply, matching the implemented weakly dispersive trend. The maximum measured phase-speed relative error is 2.45×10^{-4} for $\alpha_B = 0$ and 1.91×10^{-8} for $\alpha_B = 1/3$ in this diagnostic.

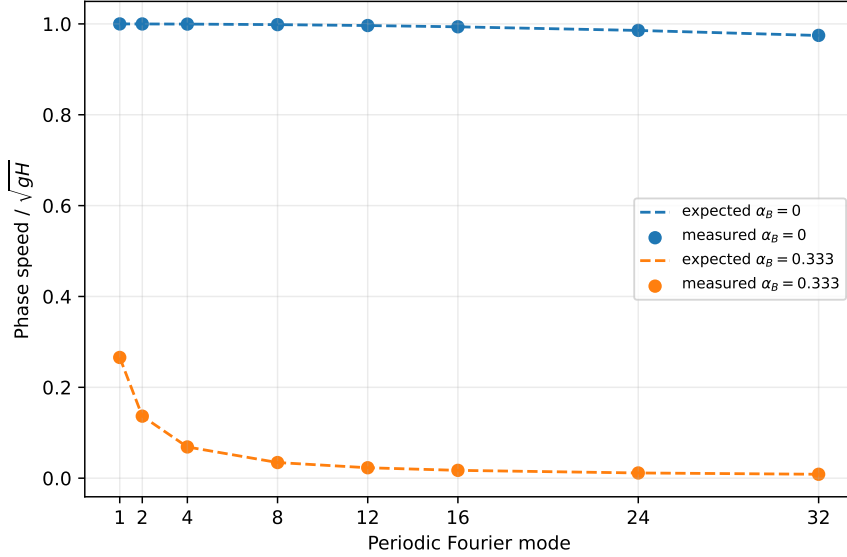


Figure 5: Appendix-only Boussinesq dispersion check on constant-depth periodic sinusoidal waves. Markers are phase speeds measured by Fourier phase tracking; dashed curves are the implemented discrete trend. This is a solver-sensitivity diagnostic, not a main result, external-solver validation, production-code validation, or event-scale validation.

Table 21: Summary of the constant-depth Boussinesq dispersion diagnostic. The $\alpha_B = 0$ row uses the direct non-CG path; the $\alpha_B = 1/3$ row uses the configured conjugate-gradient mass solve with tolerance 10^{-12} and maximum 700 iterations.

α_B	Modes	Max rel. err.	CG failed steps	CG iteration / residual summary
0	1, 2, 4, 8, 12, 16, 24, 32	2.45×10^{-4}	direct path	no CG solve
1/3	1, 2, 4, 8, 12, 16, 24, 32	1.91×10^{-8}	0	max iter. 35; max residual ratio 1.0×10^{-12}

A.3. Additional result tables

Table 22 reports the individual ensemble members used for the uncertainty study of Section 7.9. The seven members differ only in random seed and otherwise share the main FNO configuration; their best validation relative L_2 values are tightly clustered (0.240–0.245), so the ensemble spread comes from independent optimization trajectories rather than from differing model capacity or stopping criteria. Table 24 expands the runtime summary of Section 7.3 with the raw benchmark settings recorded in the per-model speed JSON and the trainable parameter counts, so the per-sample timings can be reproduced without conflating parameter count with throughput.

Table 22: Deep-ensemble members for the uncertainty study. All seven share the main hydrostatic FNO configuration and differ only in random seed. Best validation relative L_2 is the checkpoint-selection metric; epochs is the training length.

Member	Seed	Epochs	Best val. rel- L_2
1	11	202	0.2413
2	22	208	0.2454
3	33	141	0.2451
4	44	256	0.2408
5	55	168	0.2427
6	66	201	0.2409
7	77	241	0.2401

Reliability-curve diagnostic and calibration status. Figure 6 replots the full regenerated ensemble coverage summaries as a coverage-versus-nominal reliability curve. These are marginal per-element coverages over correlated pixels and frames, so the curve diagnoses interval scale but does not make an independent-sample probability claim. We fit a single post-hoc marginal scale correction, $\sigma^2 \mapsto \gamma\sigma^2$, on validation labels only, with $\gamma = 3.359364$ and corresponding standard-deviation scale 1.832857, then apply it unchanged to the test and OOD suites. Raw intervals under-cover; the scalar correction improves marginal coverage on the in-distribution test split and some OOD suites, while the hardest high source-strength tail remains under-covered. This is a post-hoc scalar correction, not operational risk calibration.

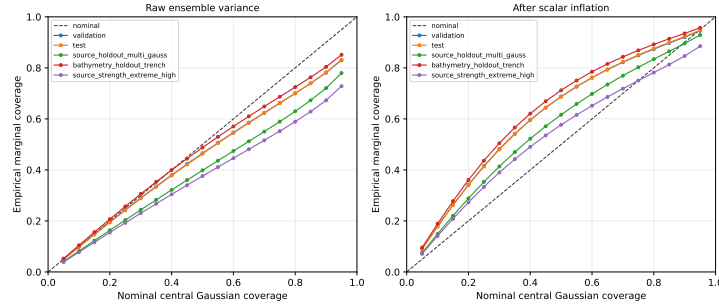


Figure 6: Full ensemble reliability curve with raw and validation-calibrated central Gaussian intervals. The ensemble undercovers before calibration; scalar variance inflation improves marginal coverage but remains a post-hoc per-element diagnostic.

Table 23: Headline nominal 90% marginal coverage before and after validation-only scalar variance inflation. Coverages are per-element over correlated pixels and frames.

Suite	Raw	Calibrated
Validation	0.7812	0.9206
Test	0.7822	0.9230
multi-gauss subset	0.7206	0.8957
trench subset	0.8039	0.9351
high source-strength tail	0.6728	0.8464

Table 24: Inference-benchmark settings from the per-model speed JSON, on the hydrostatic test split. Parameter counts are in millions. All surrogates are timed on GPU (CUDA, fp32) with batch size 64, five warm-up batches, and twenty timed repeats. Hardware: NVIDIA GeForce RTX 4050 Laptop GPU, Intel Core i5-13420H, PyTorch 2.10 / CUDA 12.8.

Model	Params (M)	Device	Precision	Batch	Warm-up / repeats	Infer (ms)
FNO	5.443	CUDA	fp32	64	5 / 20	0.904
FFNO	0.411	CUDA	fp32	64	5 / 20	1.664
CNN	0.045	CUDA	fp32	64	5 / 20	0.171
U-Net	0.104	CUDA	fp32	64	5 / 20	0.237
FNO-modes8	1.793	CUDA	fp32	64	5 / 20	0.861
FNO-modes20	11.083	CUDA	fp32	64	5 / 20	0.954
U-FNO	6.938	CUDA	fp32	64	5 / 20	3.895
WNO	2.016	CUDA	fp32	64	5 / 20	2.086

A.4. Sample-scaling diagnostic

We isolate sample-count effects by training the same hydrostatic FNO on nested random subsets of the 10,000-sample pool, with optimizer, schedule, early stopping, batch size, and test split ($N = 2500$) fixed. Figures 7–8 plot relative L_2 and companion metrics; Table 25 gives the values.

Relative L_2 drops monotonically from 0.586 at $N = 100$ to 0.235 at $N = 10,000$, with largest gains below $N = 1,000$ and diminishing returns past $N = 2,500$. This single-seed diagnostic is not a converged scaling law; the two smallest subsets ($N \leq 250$) hit the 256-epoch cap but are validation-flat to within 10^{-3} over the final 50 epochs, and larger subsets early-stop.

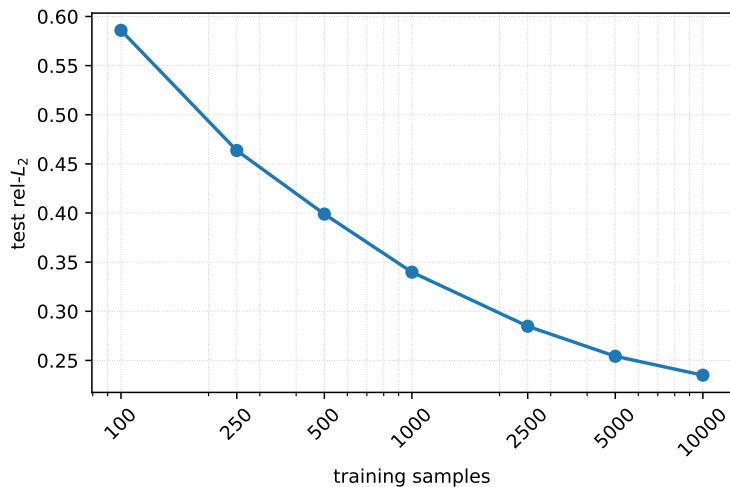


Figure 7: Single-seed sample scaling: hydrostatic-FNO test relative L_2 versus training-set size on the fixed test split ($N = 2500$).

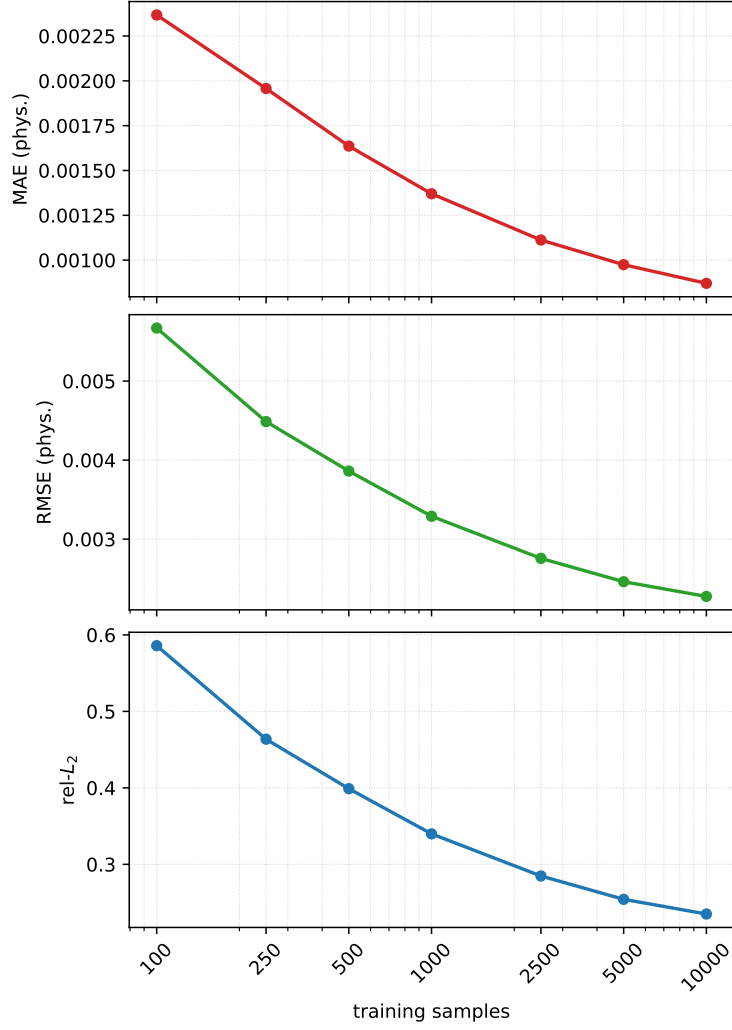


Figure 8: Companion MAE, RMSE, and relative L_2 curves for Figure 7.

Table 25: Single-seed sample-scaling diagnostic for the FNO on the hydrostatic reference. Each row is an independent training run on a nested random subset of the 10,000-sample training pool, scored on the same held-out test split ($N = 2500$) with the same architecture, optimizer, schedule, and early stopping. MAE and RMSE are in denormalized benchmark-scale surface-elevation units; relative L_2 is dimensionless; max-err is the global worst-case element in normalized units.

Train N	MAE	RMSE	rel- L_2	max-err
100	0.00237	0.00567	0.586	54.5
250	0.00196	0.00449	0.464	42.5
500	0.00164	0.00386	0.399	38.5
1,000	0.00137	0.00329	0.340	34.0
2,500	0.00111	0.00276	0.285	32.5
5,000	0.00097	0.00246	0.254	25.5
10,000	0.00087	0.00227	0.235	22.9

Table 26: Windowed rollout diagnostic details for the hydrostatic test split. Both models predict in $K = 5$ frame blocks and are first-target-frame-seeded diagnostics scored on the remaining 49 predicted target frames, not like-for-like direct 50-frame predictions.

Model	Scenarios	rel- L_2	denorm. rel- L_2	first scored	final	time (ms)
Window-FNO-5	2500	0.0771	0.0765	0.0512	0.1725	11.13
Window-FFNO-5	2500	0.0517	0.0513	0.0303	0.1695	21.63

Table 27: Window-FNO-5 on the distribution-shift and native-resolution suites, all evaluated autoregressively as first-target-frame-seeded diagnostics scored on the remaining 49 predicted target frames. The single-pass column repeats the direct FNO’s rel- L_2 on the identical suite for reference; because the windowed rollout is seeded, these are not like-for-like with direct 50-frame prediction and the windowed advantage partly reflects the easier seeded task. The native-resolution rows use the 64×64 checkpoint zero-shot: 32×32 and 128×128 are out-of-training-resolution transfers, not native-trained models. Full-traj is the rel- L_2 over scored predicted frames; final is the last-frame rel- L_2 .

Suite	Type	n	Full-traj	First scored	Final	Single-pass
multi-gauss subset	Source shift	414	0.055	0.025	0.150	0.232
trench subset	Bathymetry shift	481	0.088	0.052	0.219	0.267
high source-strength tail	Strength tail	180	0.092	0.045	0.221	0.247–0.345
32×32 (zero-shot)	Native res.	150	1.101	0.455	5.416	—
64×64 (training res.)	Native res.	150	0.076	0.056	0.148	—
128×128 (zero-shot)	Native res.	150	0.828	0.258	0.985	—

A.5. Real-bathymetry diagnostics

Table 28: Status of the real-bathymetry diagnostics. The main paper uses the fully wet 10-crop morphology suite; the single Tohoku crop remains an illustrative figure case. The coastline crop is evaluated both as a fully wet ablation and as a wet-dry OOD stress test.

Crop	Rescale	Status	Reading
10-crop offshore suite	$[-10, -0.75]$	Main	Fully wet morphology transfer
Tohoku offshore	$[-10, -0.75]$	Figure	Illustrative suite crop
Coastline	$[-10, -0.75]$	Appendix	Geometry ablation
Coastline	$[-10, 5]$	Appendix	Wet-dry stress

Table 29 lists the ten crops behind the aggregate real-bathymetry row in Table 11.

Table 29: Per-crop 10-crop fully wet real-bathymetry morphology suite. Values are per-crop normalized trajectory rel- L_2 under synthetic forcing. FNO/FFNO are direct columns; W-* columns are first-target-frame-seeded diagnostics scored on the remaining 49 predicted target frames, so they are not like-for-like with direct 50-frame prediction. Bounds are W/E/S/N.

Crop	Bounds	FNO	FFNO	W-FNO-5	W-FFNO-5
Sumatra-Java trench	95/110/-20/-5	0.211	0.120	0.052	0.039
Alaska-Aleutian margin	-155/-140/45/60	0.286	0.161	0.095	0.091
Cascadia shelf-slope	-135/-123/40/52	0.235	0.267	0.048	0.030
Hellenic Arc/Aegean	25/30/32/37	0.285	0.227	0.037	0.037
Florida-Bahamas shelf	-80/-65/20/35	0.252	0.142	0.077	0.064
Caribbean island arc	-70/-60/10/20	0.202	0.156	0.073	0.059
Philippines east trench	125/135/5/15	0.219	0.110	0.051	0.028
Kuril-Japan trench	146/160/36/50	0.510	0.288	0.064	0.061
Japan Trench/Tohoku offshore	141/147/34/40	0.397	0.186	0.051	0.035
Nankai Trough/southwest Japan	138/144/30/36	0.530	0.439	0.048	0.048

Table 30: Direct and windowed models on the synthetic-source real-bathymetry diagnostics. Values are dimensionless relative L_2 . Windowed rows are first-target-frame-seeded diagnostics scored on the remaining 49 predicted target frames and are not like-for-like with direct 50-frame prediction. The coastline wet-dry row is outside the fully wet training distribution and is included only as an appendix stress test.

Crop	n	Direct FNO	Direct FFNO	Window-FNO-5	Window-FFNO-5
10-crop offshore suite	10	0.275	0.183	0.053	0.039
Tohoku offshore	1	0.228	0.165	0.046	0.029
Coast wet	1	0.248	0.128	0.065	0.034
Coast wet-dry	1	0.996	0.986	0.981	0.973

Both coastline rows use the same GEBCO-derived Japan-Trench-region crop (internal tag `gebco_japan_coast`), covering lon 138°–144°E and lat 30°–36°N, resampled to 64×64 under the identical Okada-like synthetic source (strength 0.826, seed 367) as the illustrative offshore diagnostic; only the depth-range rescale differs. The crop is taken from the GEBCO_2026 Grid [7]. The two coastline rows separate morphology from wet-dry shift. When the coastline geometry is linearly rescaled into the fully wet range $[-10, -0.75]$, all four model variants remain usable. When the same coastline crop keeps positive topography in the $[-10, 5]$ range, the reference rollout triggers a small quality warning (maximum $|\eta| = 5.0135$ against the configured limit 5.0), and all variants rise to rel- $L_2 \approx 1.0$. We therefore interpret the stress failure mainly as a dry-land/positive-topography distribution shift, not as evidence that coastline geometry alone breaks the surrogate.

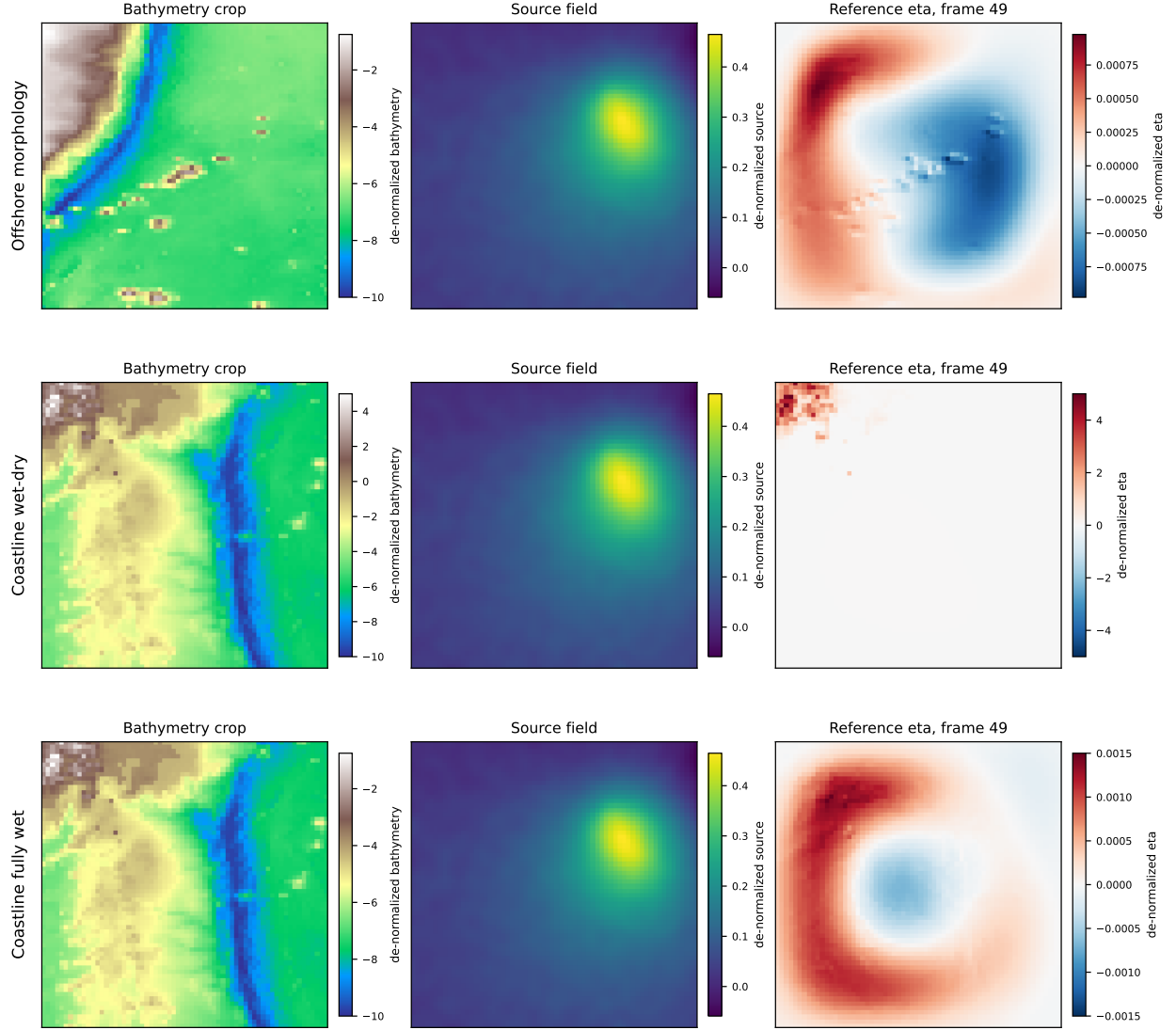


Figure 9: Real-bathymetry appendix suites under the identical synthetic Okada-like source. One row per suite: de-normalized bathymetry crop, source field, and hydrostatic reference surface elevation η at frame 49. Top: offshore morphology crop, fully wet. Middle: coastline crop with positive topography retained (wet-dry stress test); the reference η develops large-amplitude responses over the dry-land region that drive the $\text{rel-}L_2 \approx 1.0$ failure in Table 30. Bottom: the same coastline geometry forced fully wet, which all reported model variants handle.

A.6. Selected high-error failure cases

Figure 10 collects four selected high-error examples rather than average or representative cases. The first row is a rough-source hard case from the ordinary hydrostatic test split (`sample_000385`, de-normalized $\text{rel-}L_2 = 0.598$ for the main FNO). The second is a strict held-out rough-source failure (`sample_001861`, de-normalized $\text{rel-}L_2 = 0.795$ for the rough-holdout FNO). The third is

the Nankai Trough / southwest Japan crop, the worst direct-FNO case in the fully wet 10-crop real-bathymetry morphology suite (normalized $\text{rel-}L_2 = 0.530$, de-normalized $\text{rel-}L_2 = 0.510$). The fourth is the coastline wet-dry stress failure (sample_000001, de-normalized $\text{rel-}L_2 = 0.996$), whose raw reference rollout carries the warning $\text{max_abs_eta}(5.01349) > \text{max_abs_eta_limit}(5)$. Within each row, the reference and prediction share the same diverging η scale and the absolute-error panel uses a separate sequential scale.

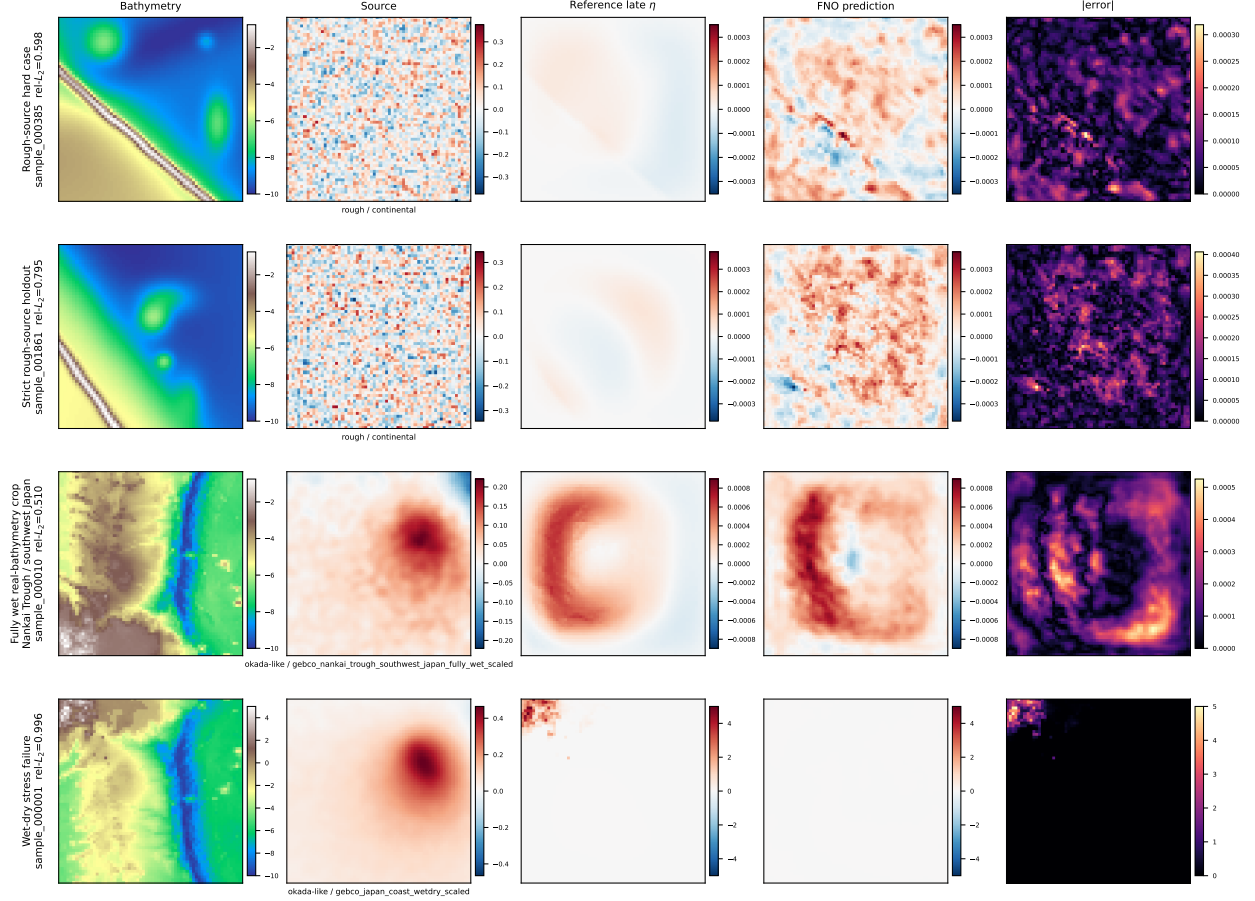


Figure 10: Selected high-error cases, not average or representative cases. Columns show de-normalized bathymetry, source, late-frame hydrostatic reference η , model prediction, and absolute error. Rows cover an ordinary rough-source hard case, a strict held-out rough-source failure, the worst fully wet real-bathymetry crop, and the wet-dry coastline stress failure.

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